

The Value of Visualization

Sarah Groves

Learning Objectives

The historical value of visualization

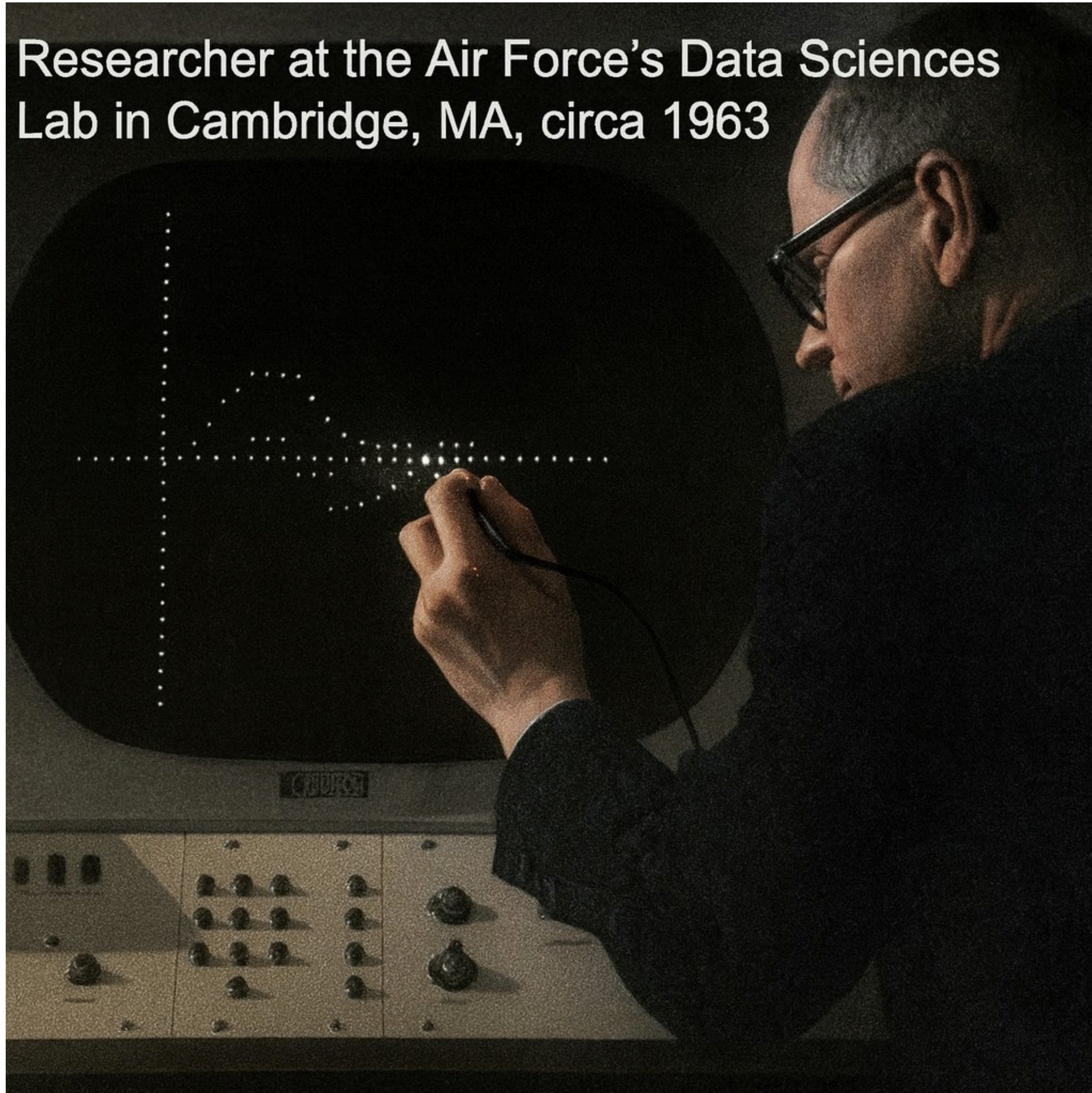
Distinguish between the different purposes of data visualization.

Learn about the advantages of visualization.

Recognize when data visualizations can help solve problems.

The First Data Scientists

Researcher at the Air Force's Data Sciences Lab in Cambridge, MA, circa 1963

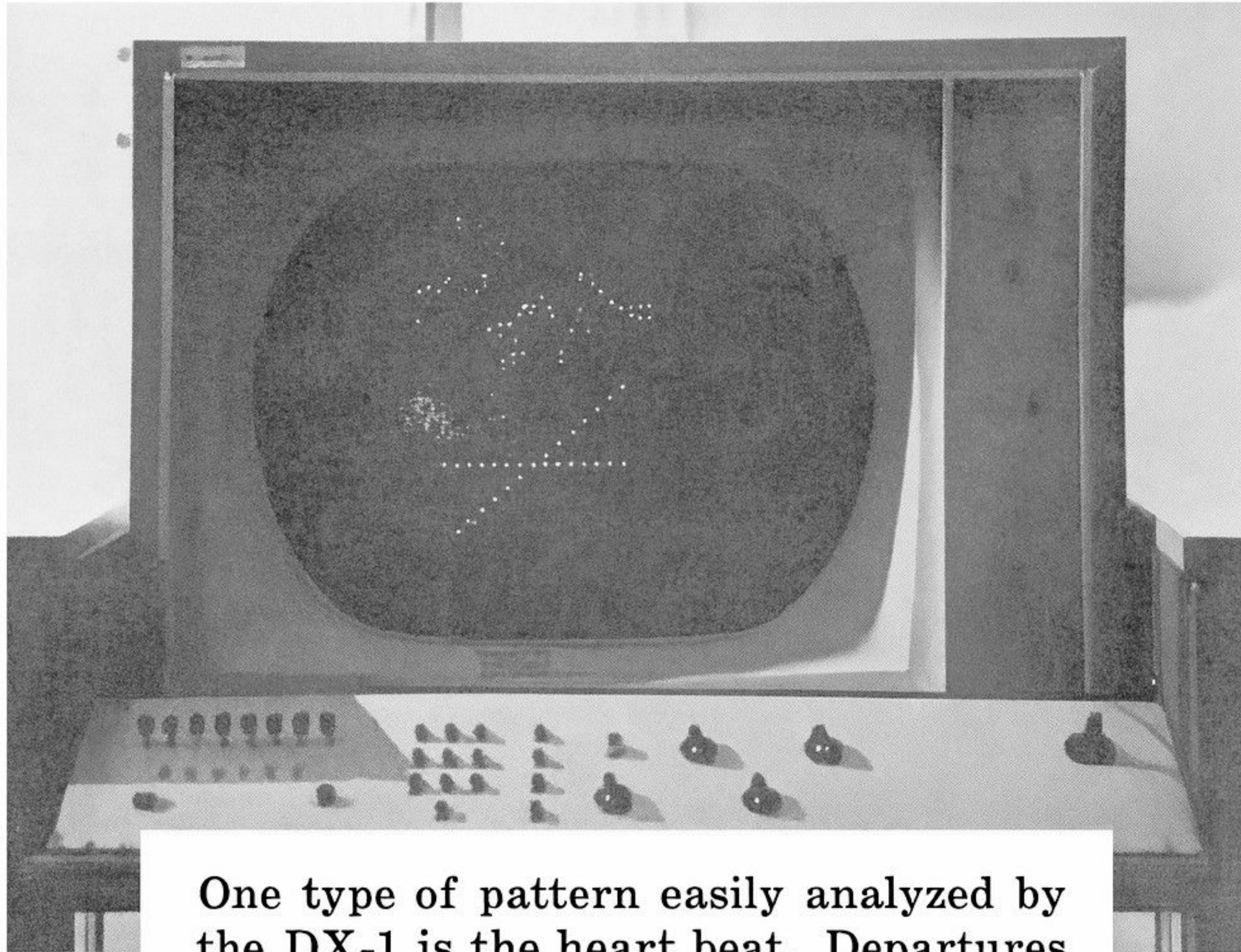


Data Science originated in the 1960s in the **military-industrial** sector

The **Data Sciences Lab** (1963) applied statistical and **computational methods** to the **data deluge** from electronic **communication** and advanced scientific **instruments**

It moved into the **sciences** in following decades and into **tech** in 2008

Humans in the Loop



One type of pattern easily analyzed by the DX-1 is the heart beat. Departures from the normal cardiogram waveform are immediately recognized by the data processor. The patterns of several successive heartbeats are shown on the scope.

Data science has always used **computational visualization** as an **analytic tool**

Pictured on the left is the DSL's **DX-1, an engineering design machine (EDM)** that used a light pen for graphical input

It was designed to **center the role of human visualization** in work of data processing

The DX-1 allowed operators
to **formulate**
engineering problems
graphically



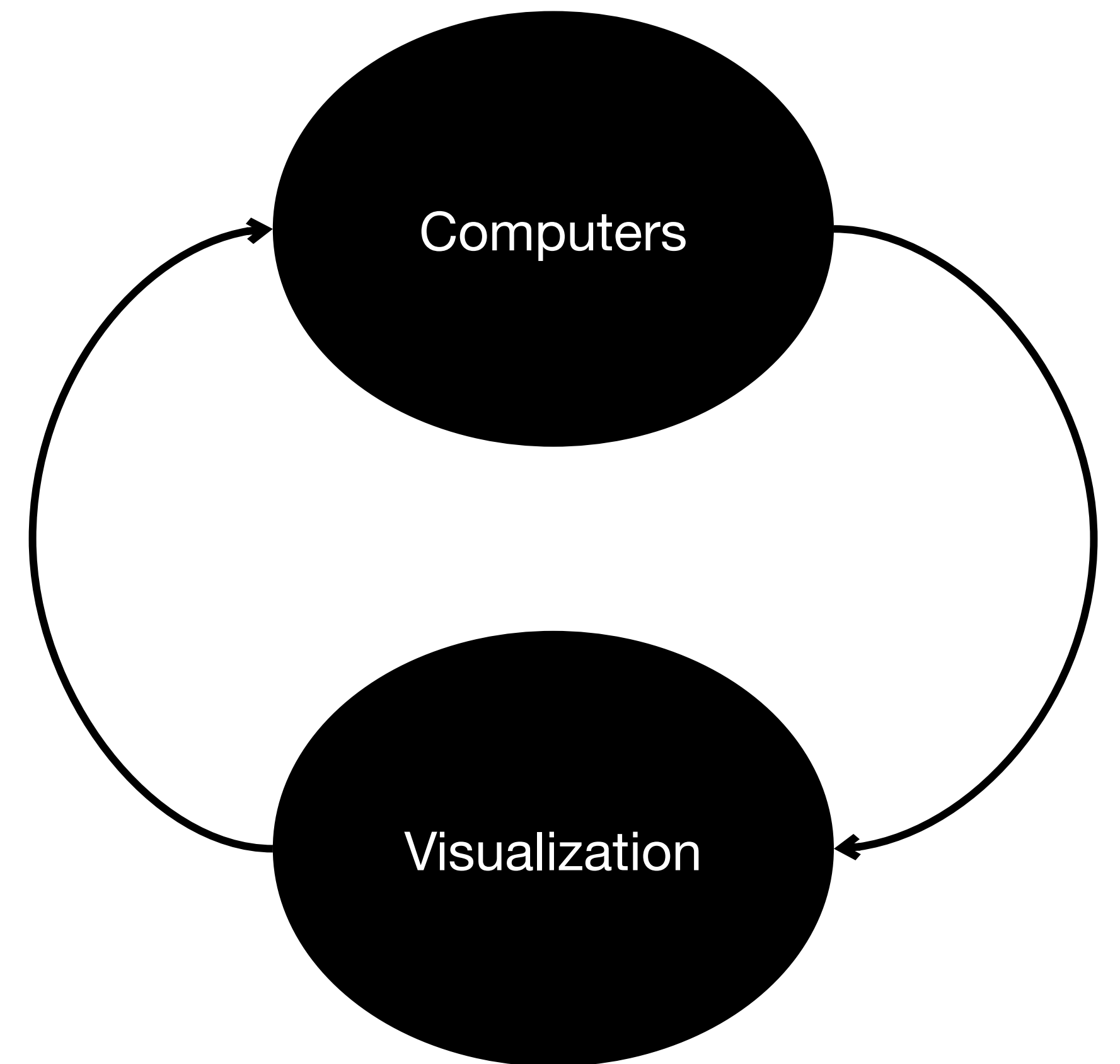
“The greatest value of a picture is when it forces us to notice what we never expected to see.”

John W. Tukey

Exploratory Data Analysis (1977)

More computation = more data = more visualization

Naturally, as computer analysis became more common, data visualization had to follow



Why Use Computers?

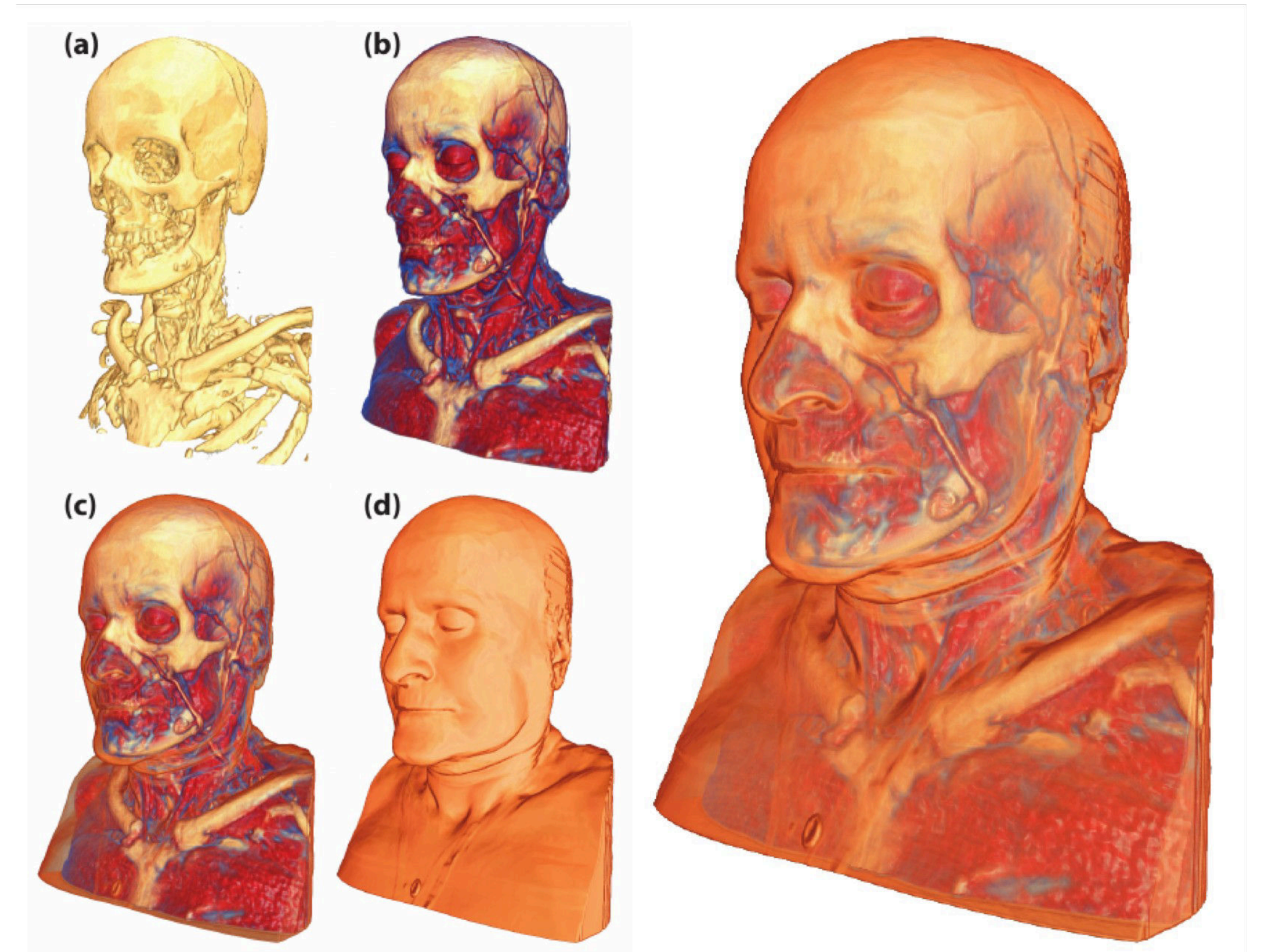
Scale

Drawing by hand (or Illustrator)

infeasible

inflexible (updates!)

How to draw an MRI scan?



[Bruckner 2007]

Why Use Computers?

Interaction

Interaction allows to “drill down” into data

Integration

Integration with algorithms

Make visualization part of a data analysis pipeline

Efficiency

Re-use charts / methods for different datasets

Quality

Precise data driven rendering

Storytelling

Use time

Why data visualization?

Write down at least 5 reasons to
visualize things.

The Value of Visualization

Aka why create visualizations?

Record information

Blueprints, photographs, seismographs, ...

Expand memory

Analyze data to support reasoning (exploratory visualization)

Develop and assess hypotheses

Find patterns / Discover errors in data

Gain intuition

Communicate information to others (explanatory visualization)

Share and persuade

Collaborate and revise

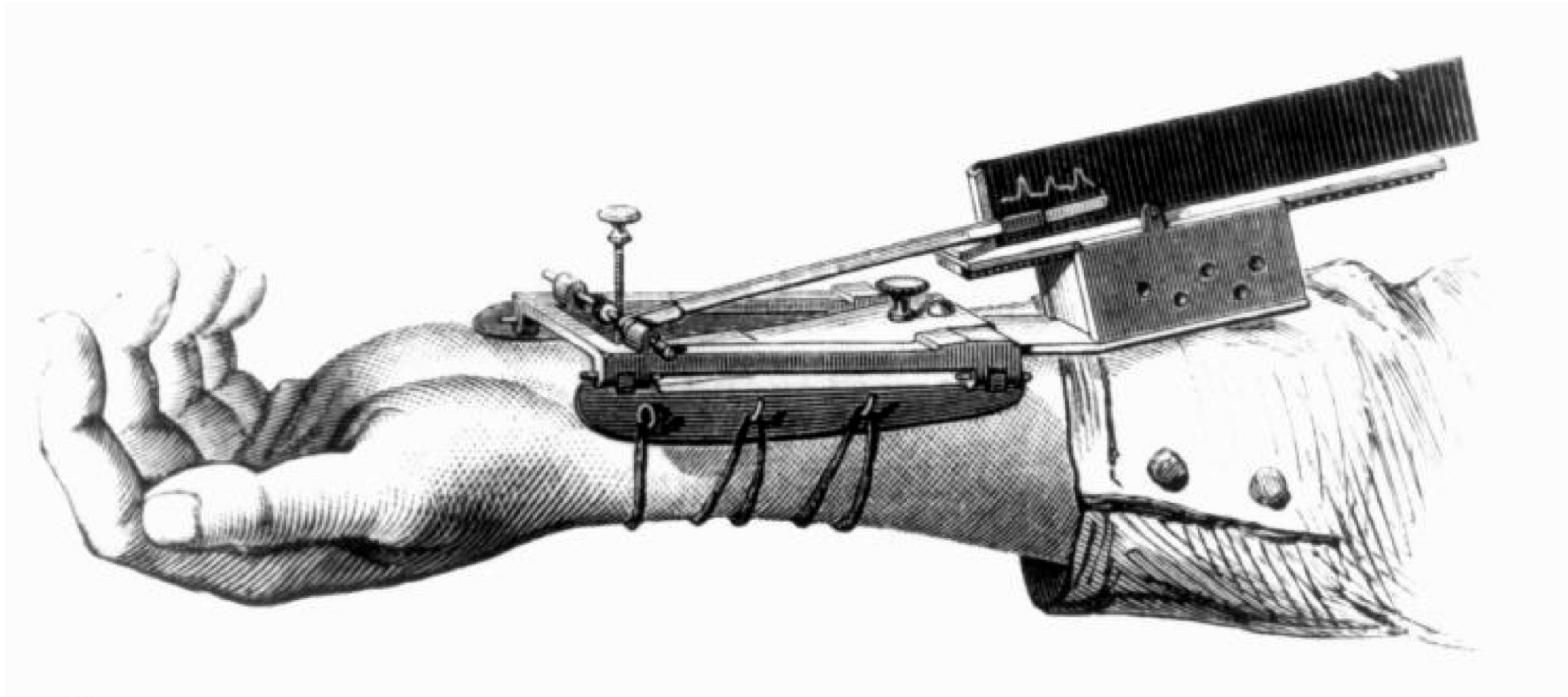
"Vision trumps all other senses. We learn and remember best through pictures, not through written or spoken words."

— John Medina, 2008, *Brain Rules*

Record Information

E.J. Marey's sphygmograph (1863) Directly recorded pulse to measure blood pressure.

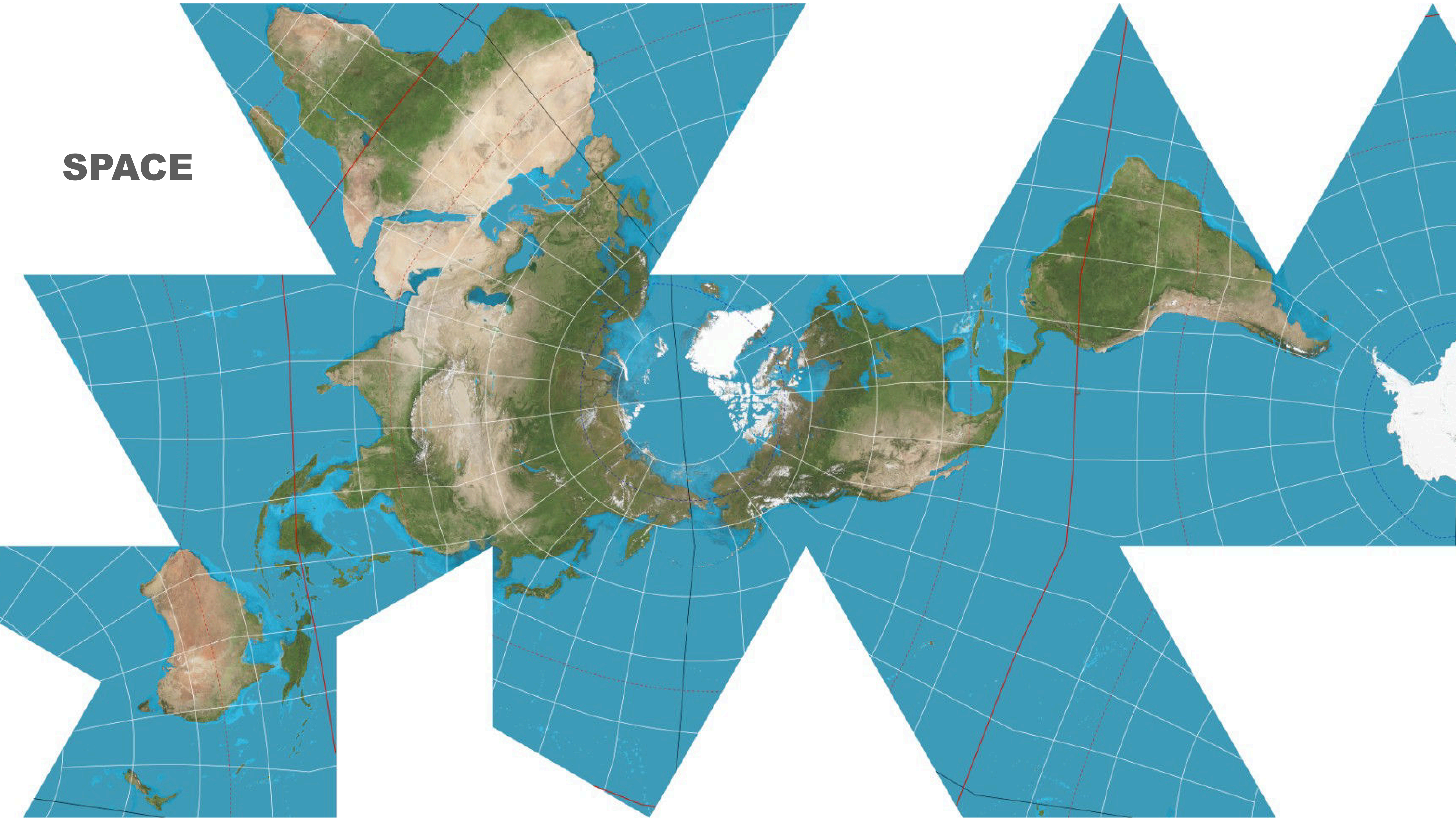
First external, non-intrusive way as a waveform.



Expanding memory:

Visualization allows us to see things
that are too big (or small) to see

SPACE

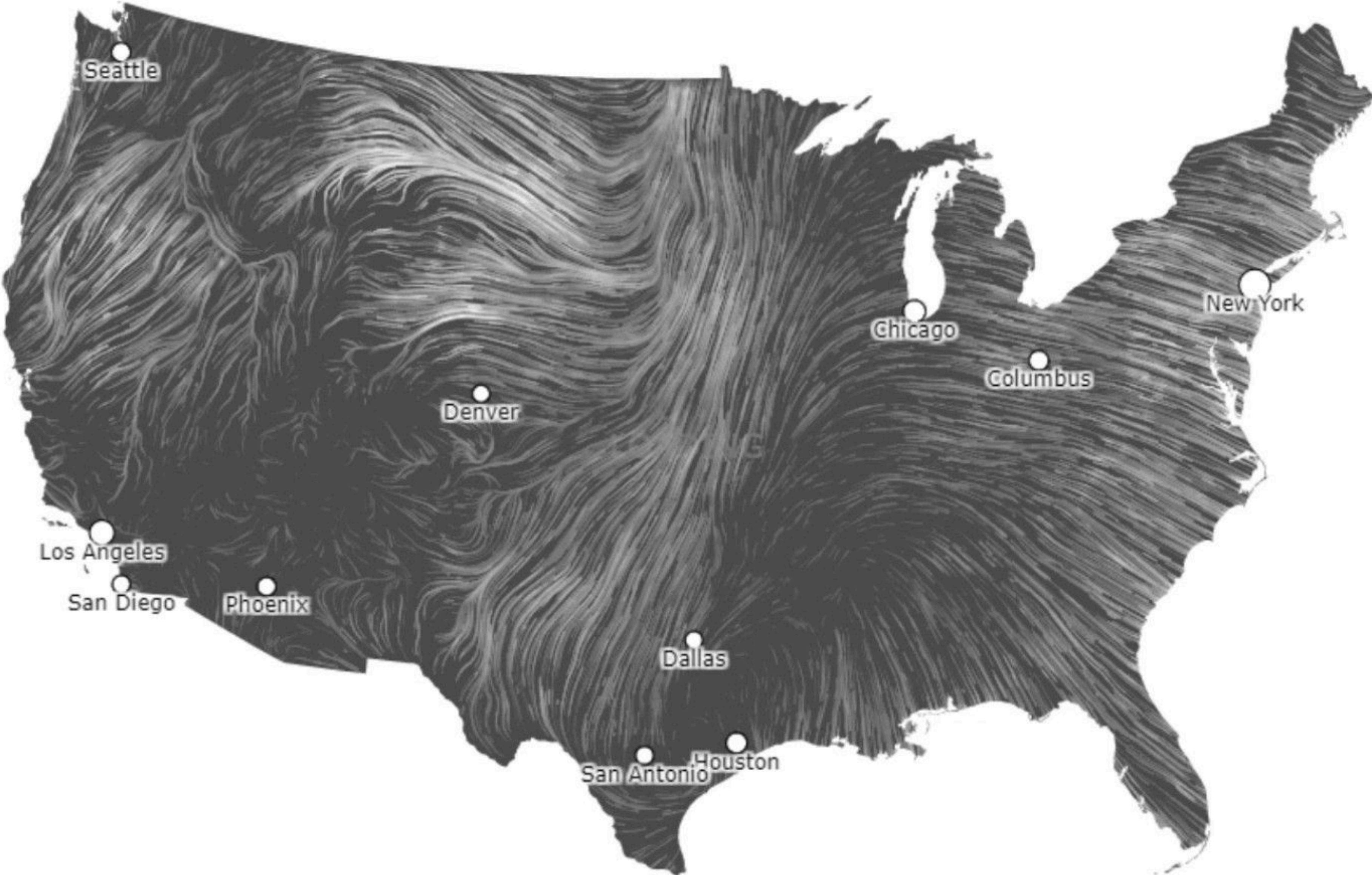
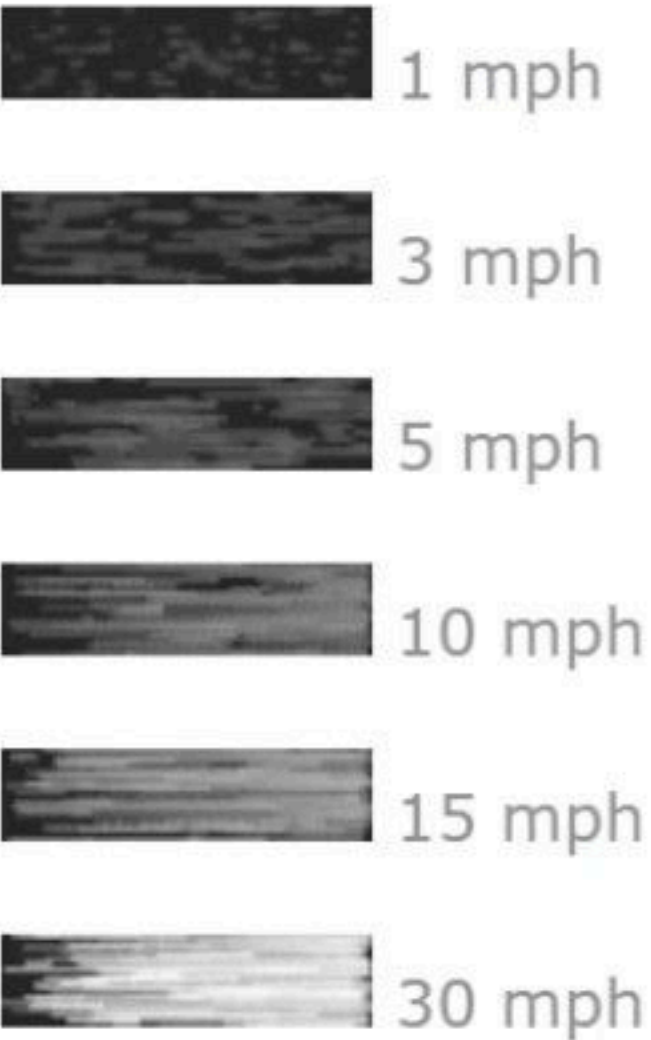


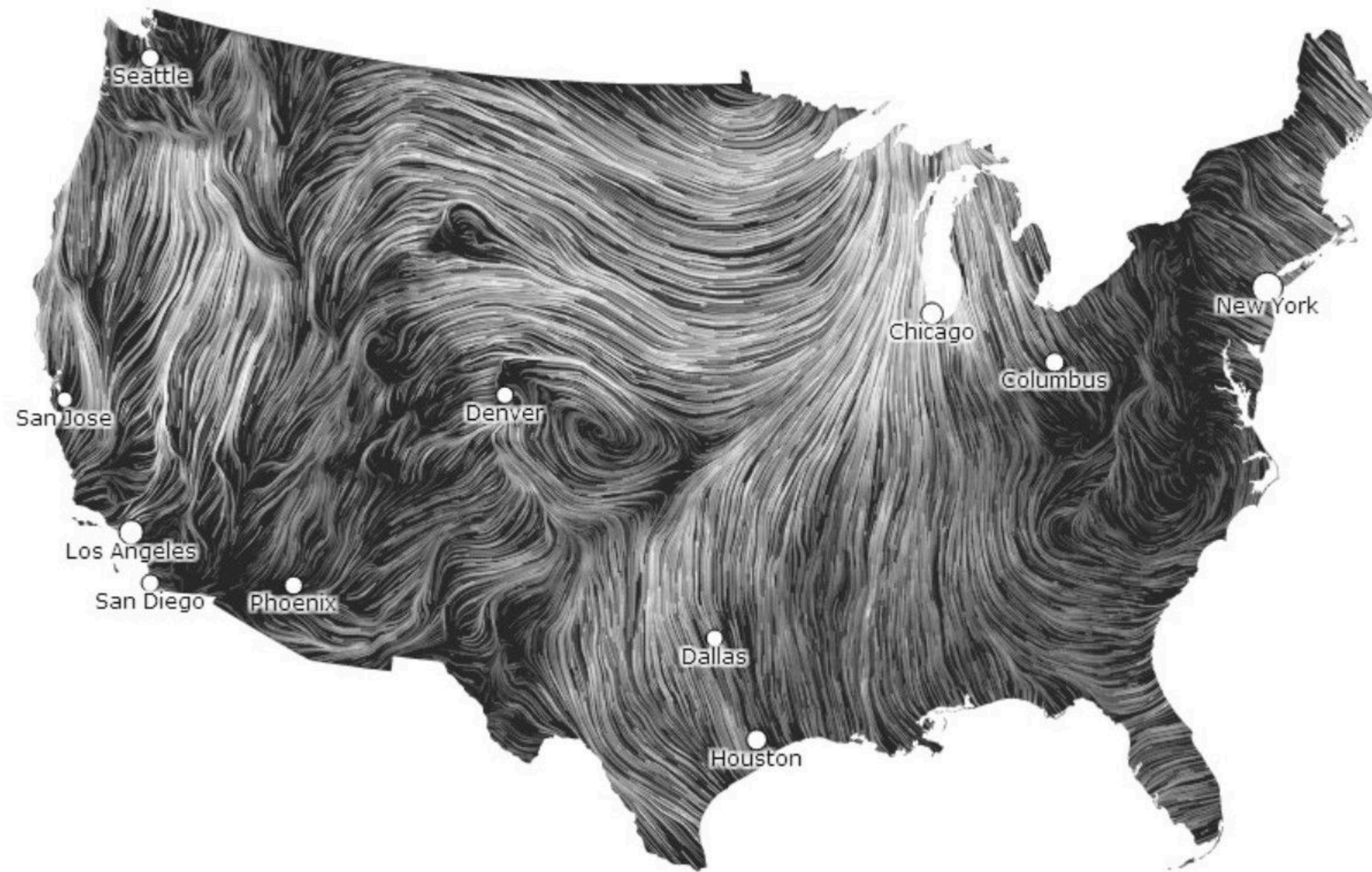
March 12, 2022

5:12 pm EST

(time of forecast download)

top speed: 50.5 mph
average: 14.4 mph

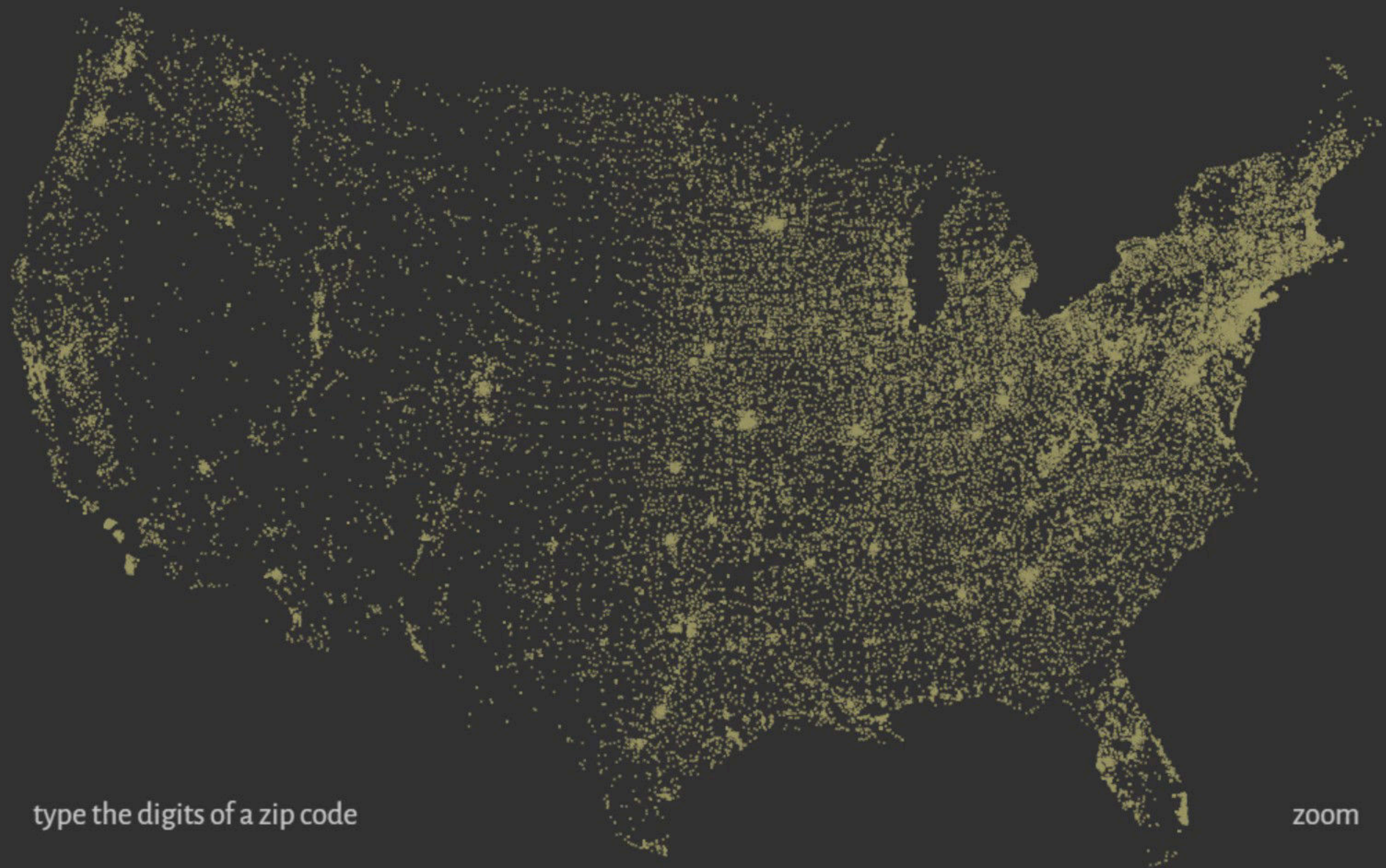




Fernanda Bertini Viégas, Martin Wattenberg

Wind Map

2012



type the digits of a zip code

zoom

Analyze Data: Exploratory visualization

- Develop and assess hypotheses
- Find patterns / Discover errors in data

Develop and assess hypotheses

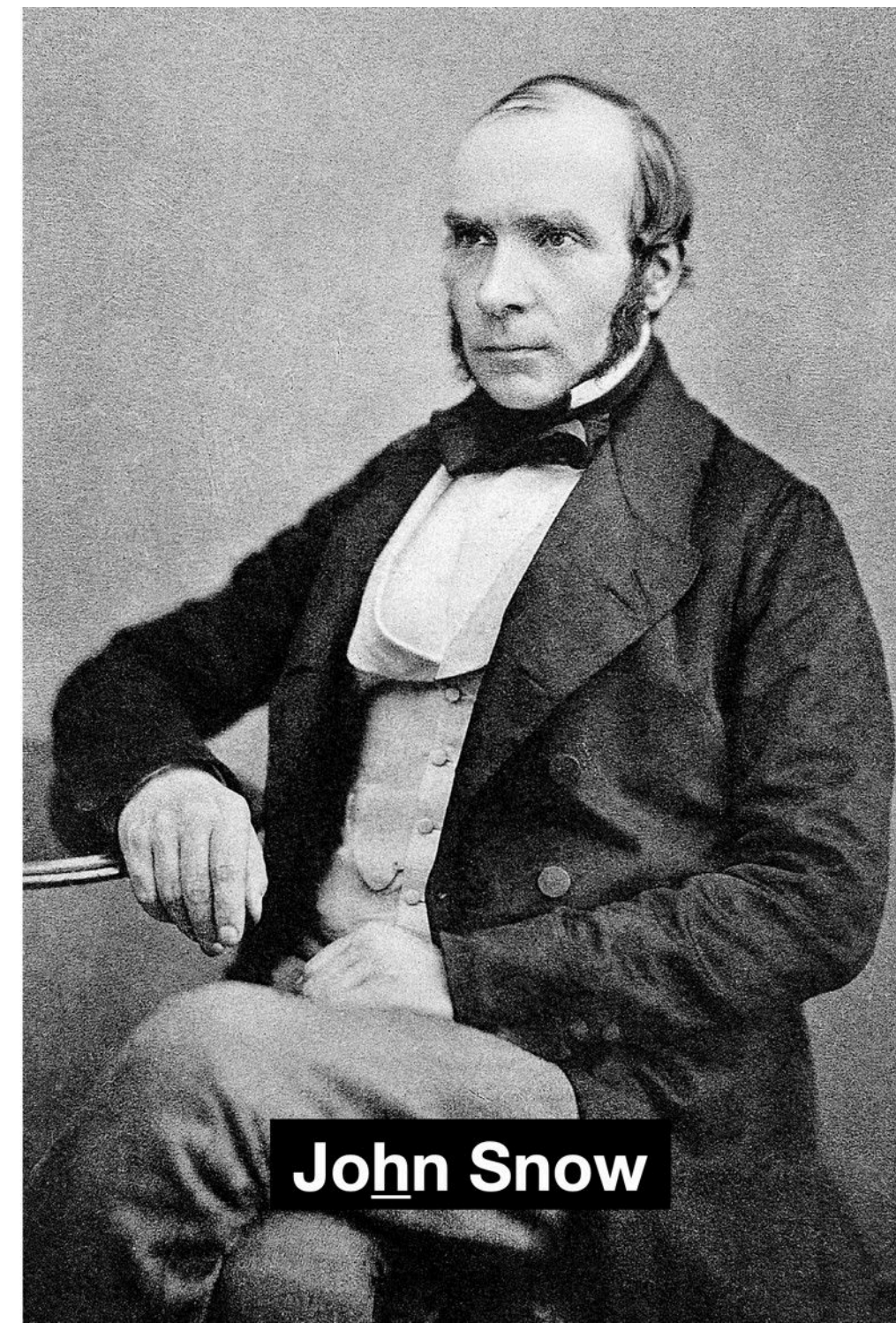
Data in Context

Cholera Outbreak

In 1854 John Snow plotted the position of each cholera case on a map.



Jon Snow



John Snow



Data in Context

Cholera Outbreak

Used map to hypothesize that pump on Broad St. was the cause.



Finding patterns in the data:
consider the following example

Country	Population (millions)	GDP (trillions \$)
USA	331	23.0
China	1441	17.7
Japan	126	4.9
Germany	83	4.2
India	1380	3.1
Brazil	213	1.6

Take 30 seconds to take in this data set

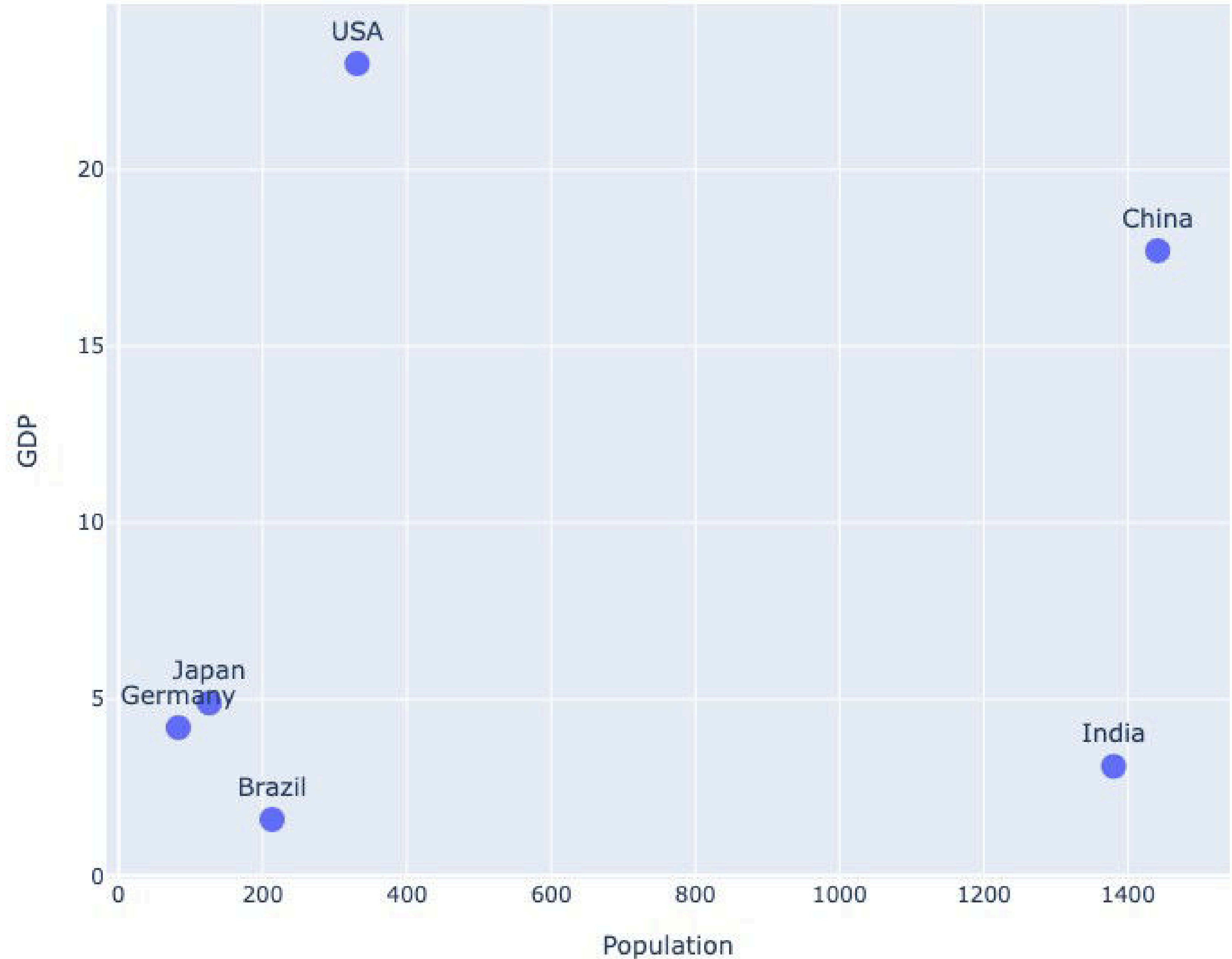
You will be asked some questions about it after a minute

Which country had the largest GDP?

Which two countries had nearly equal GDPs?

Roughly how many millions of people live in
Brazil?

Population vs GDP by Country



Which two countries were similar in population and GDP?

Which country had a large GDP for its population?

Which country had a low GDP for its population?

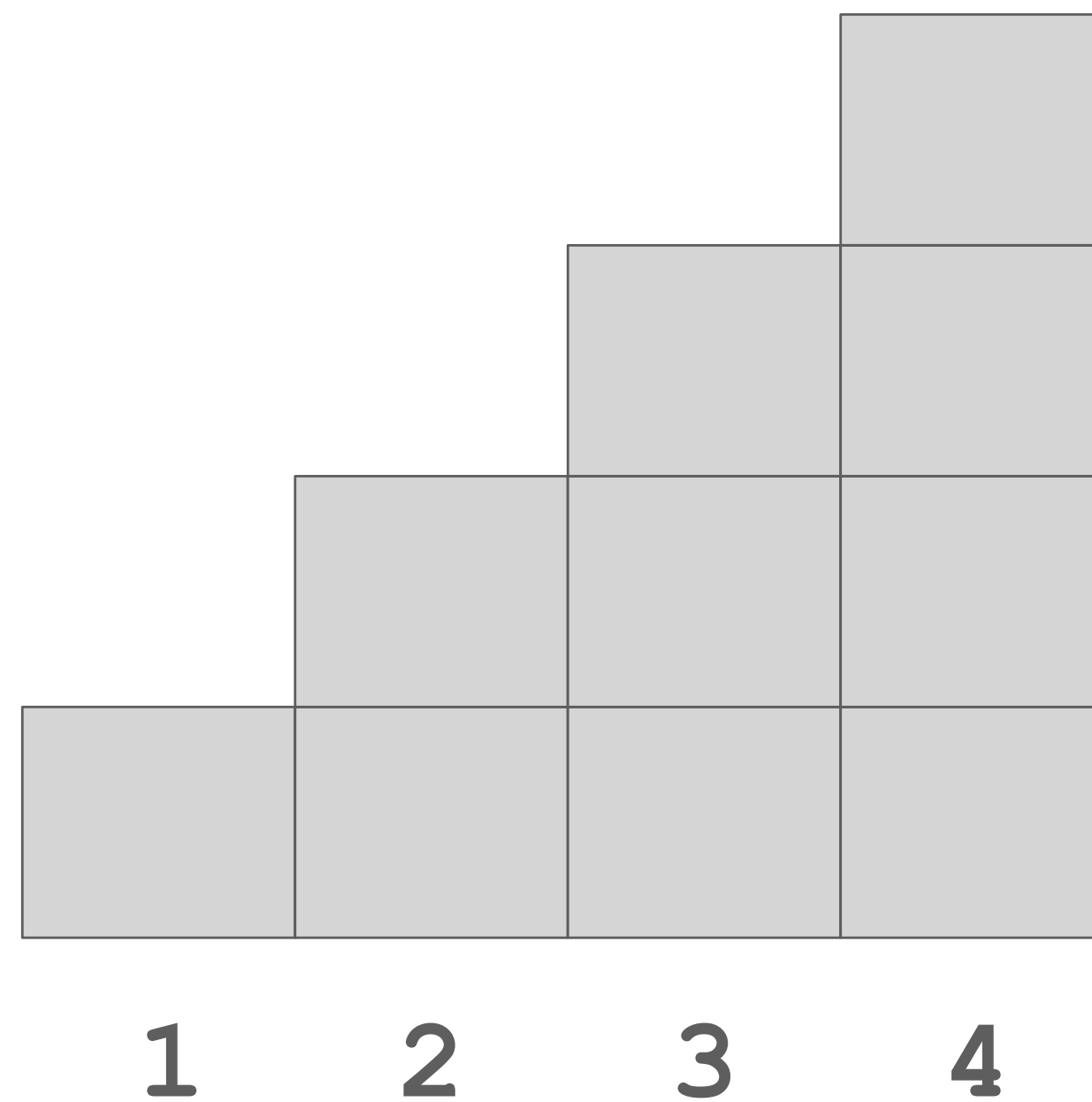
Gaining intuition:

Mathematical intuition is often visual

What is

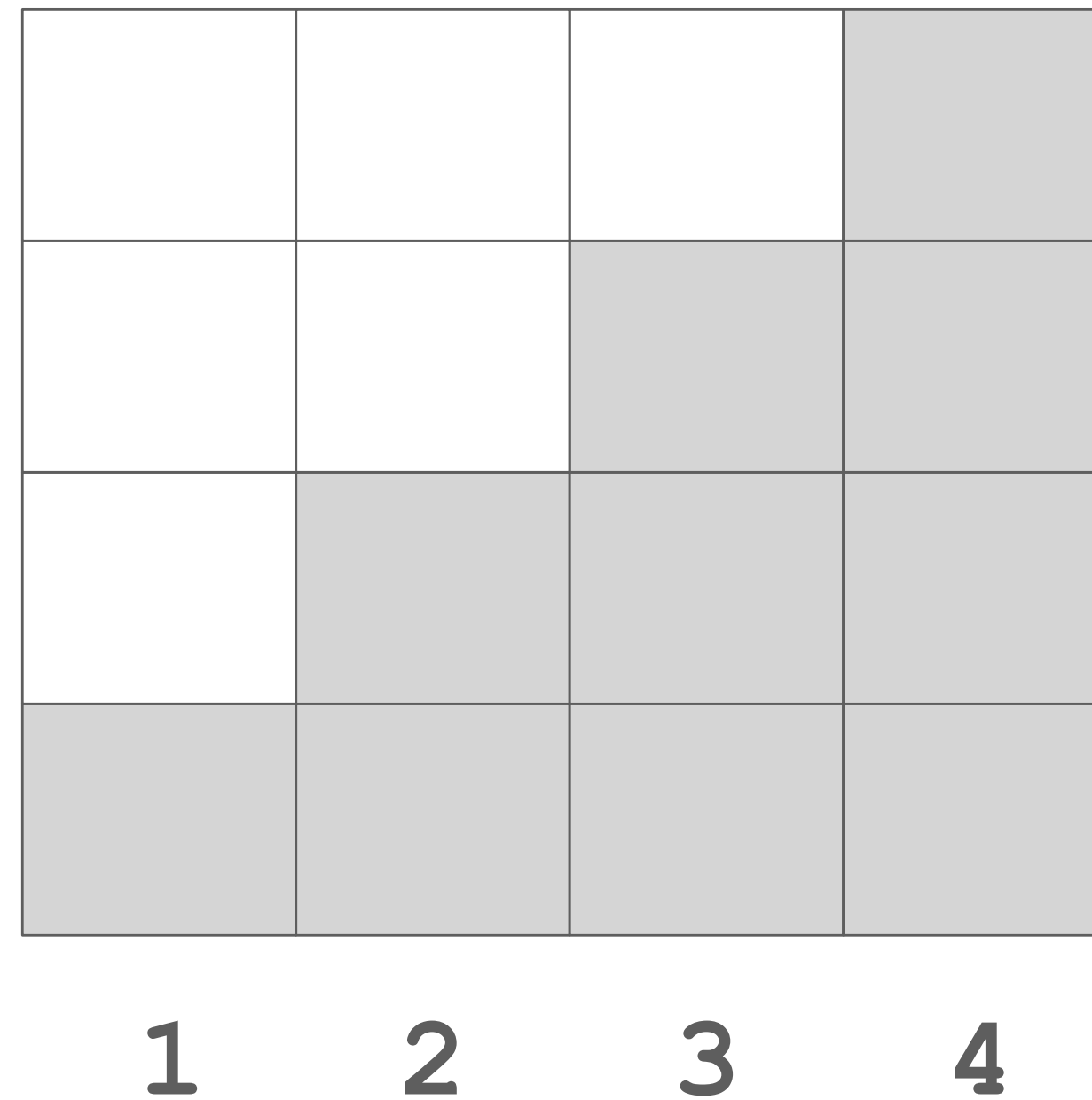
$$1 + 2 + 3 + \dots + 100 \text{ ?}$$

Let's visualize a simple example



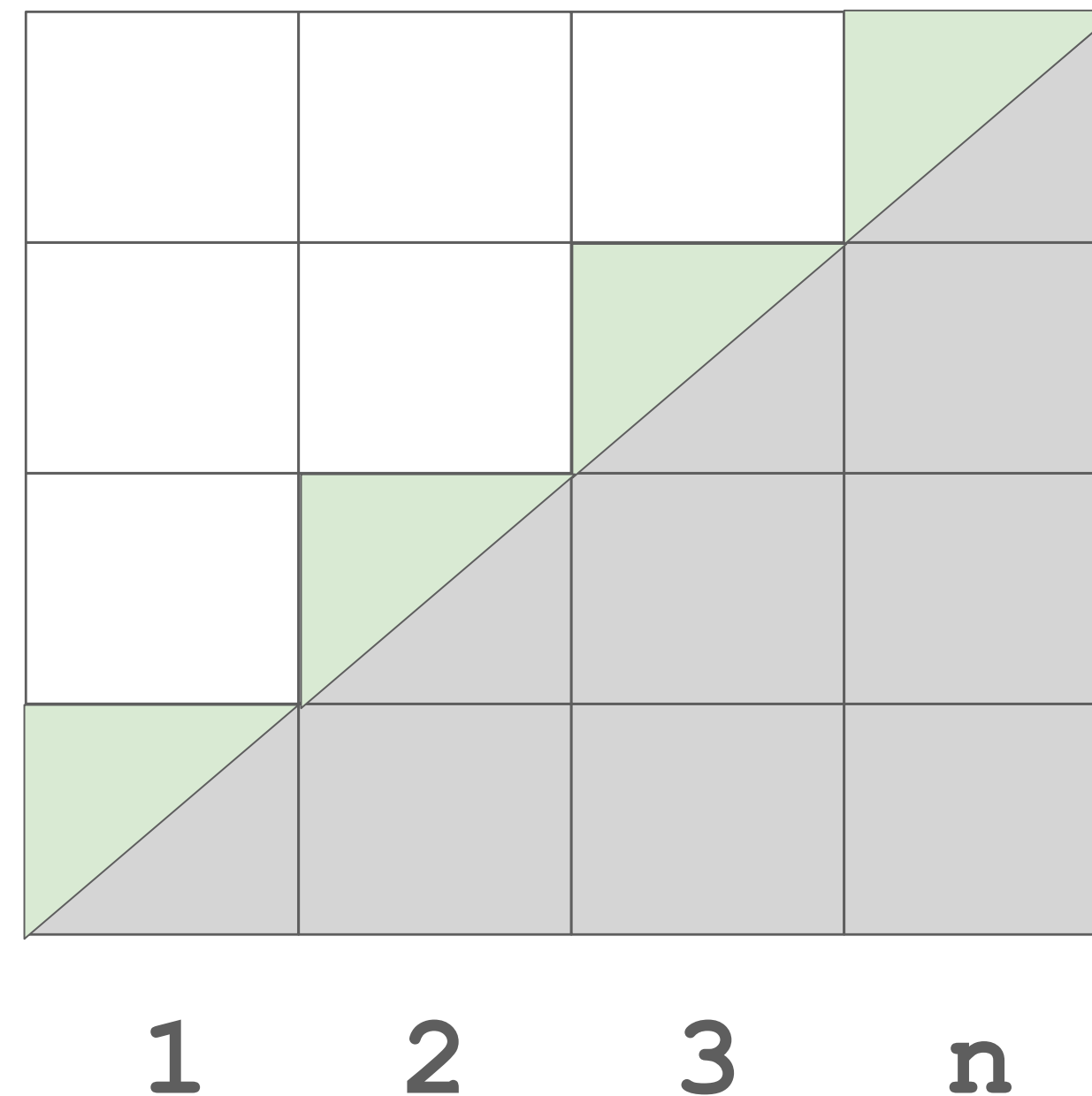
= 10

Looks like the answer is related to the square of
4 . . .



It looks something like half of the square

Let's unpack this intuition visually



$$= \frac{n^2}{2} + \frac{n}{2}$$



$$\frac{n^2 + n}{2}$$

“Mathematics is, in the end, grounded not in logic but in **the embodied capacity of human beings to imagine, to visualize, and to manipulate mental images.**”

George Lakoff & Rafael Núñez *Where Mathematics Comes From* (2000)

Finding Patterns:

Visualization exposes structure

Set 1		Set 2		Set 3		Set 4	
X	Y	X	Y	X	Y	X	Y
10	8.04	10	9.14	10	7.46	8	6.58
8	6.95	8	8.14	8	6.77	8	5.76
13	7.58	13	8.74	13	12.74	8	7.71
9	8.81	9	8.77	9	7.11	8	8.84
11	8.33	11	9.26	11	7.81	8	8.47
14	9.96	14	8.1	14	8.84	8	7.04
6	7.24	6	6.13	6	6.08	8	5.25
4	4.26	4	3.1	4	5.39	19	12.5
12	10.84	12	9.11	12	8.15	8	5.56
7	4.82	7	7.26	7	6.42	8	7.91
5	5.68	5	4.74	5	5.73	8	6.89

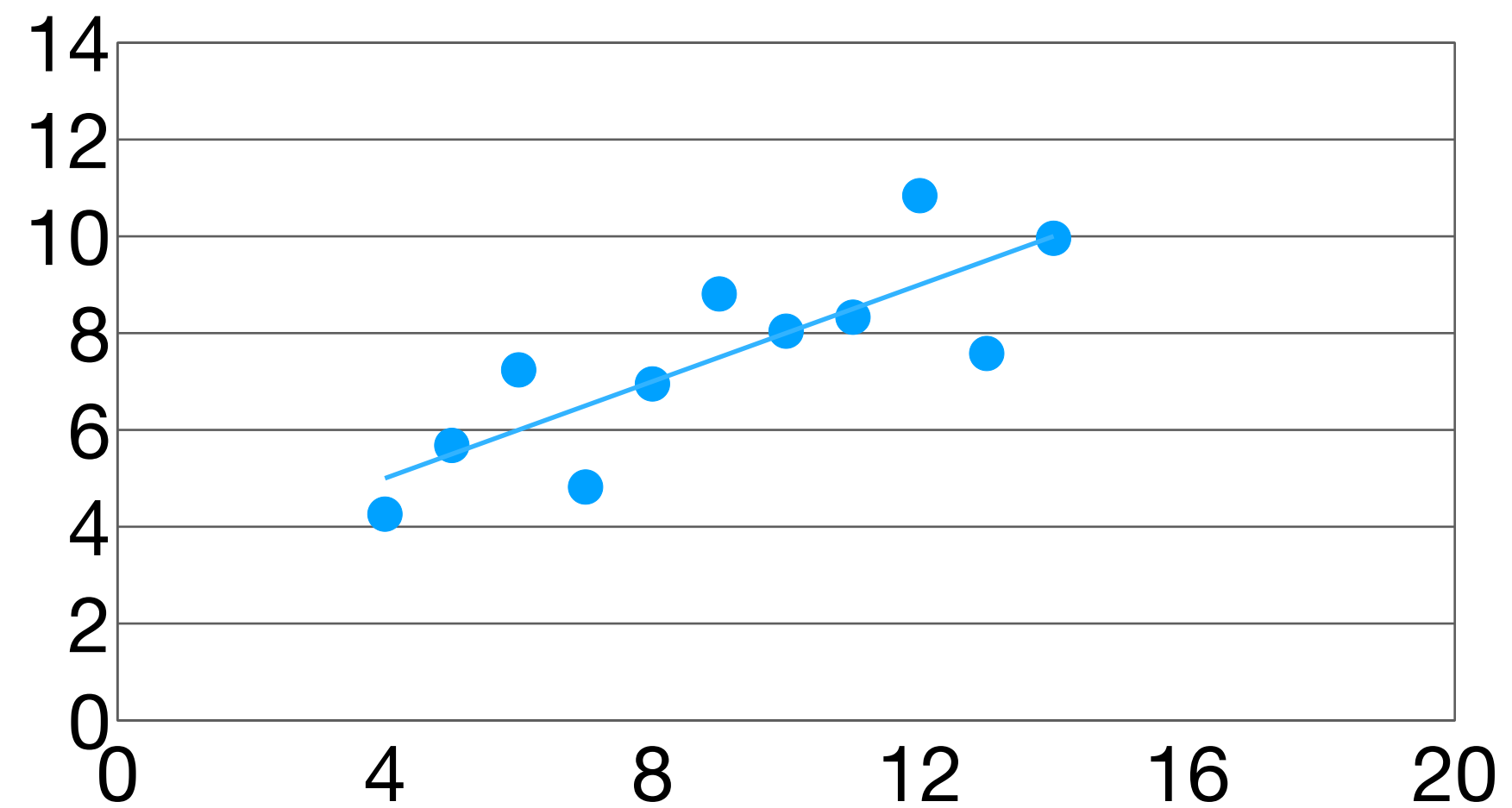
Summary Statistics Linear Regression

$u_X = 9.0$ $\sigma_X = 3.317$ $Y^2 = 3 + 0.5 X$

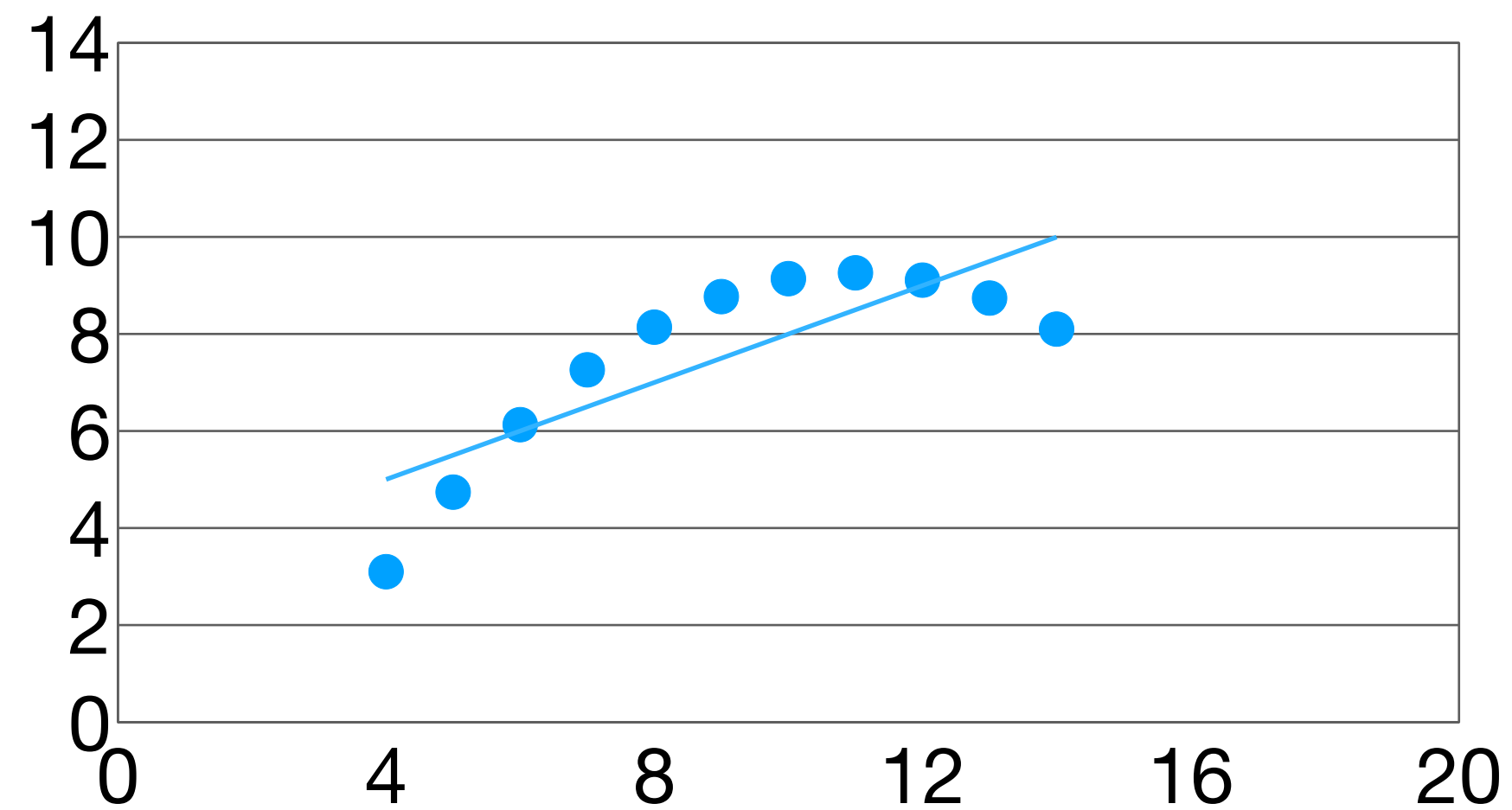
$u_Y = 7.5$ $\sigma_Y = 2.03$ $R^2 = 0.67$

[Anscombe 1973]

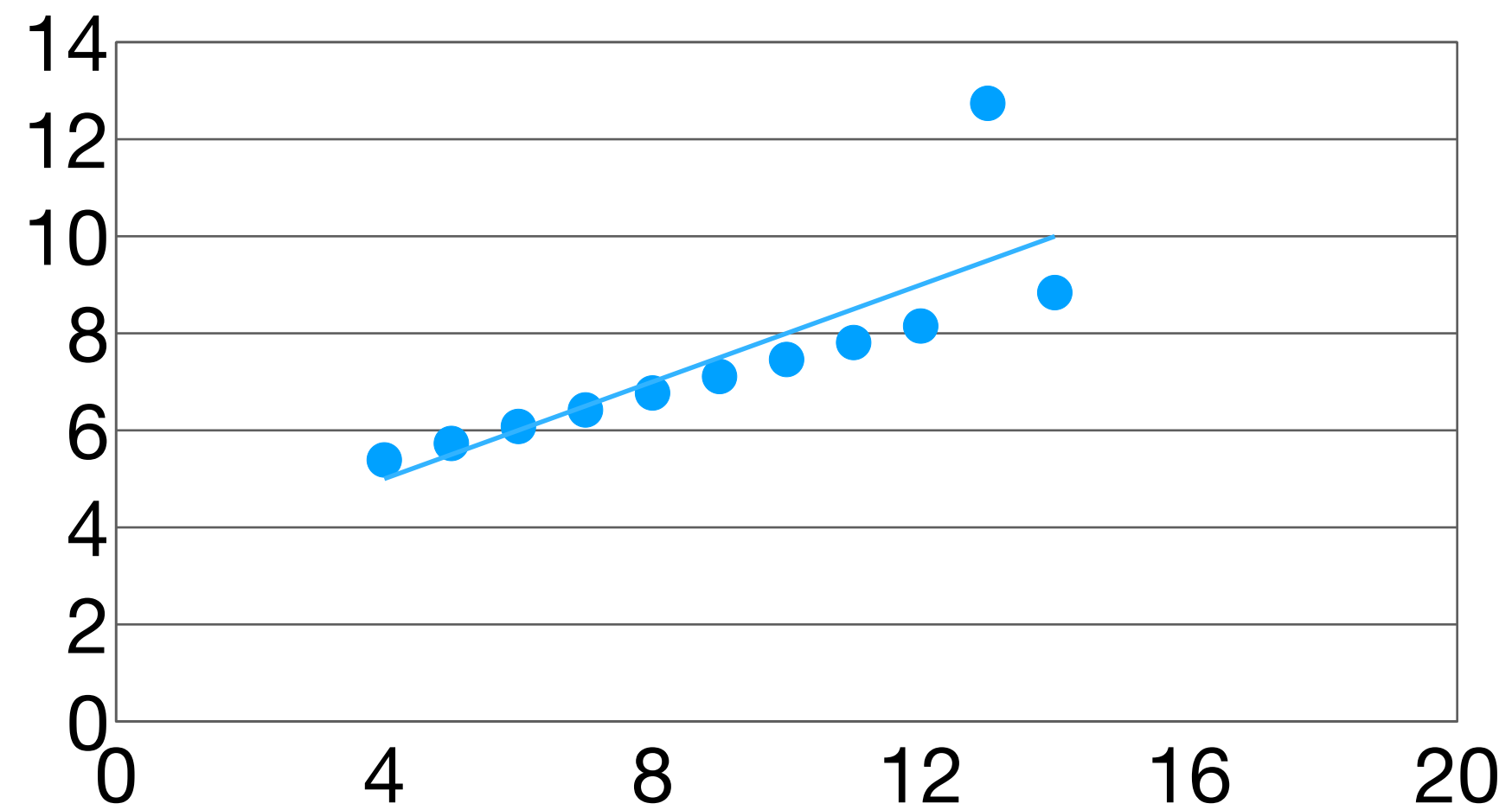
Set 1



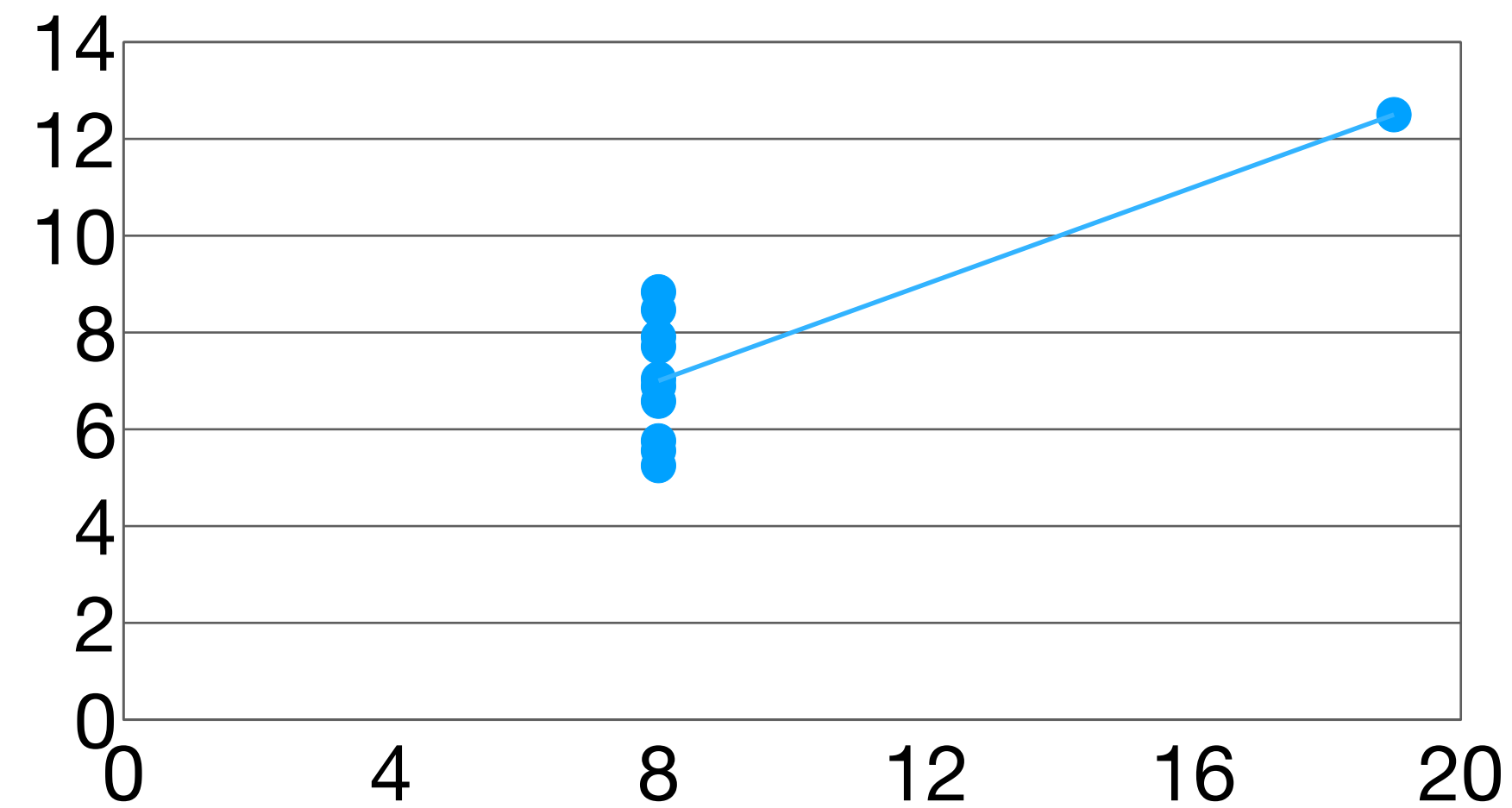
Set 2

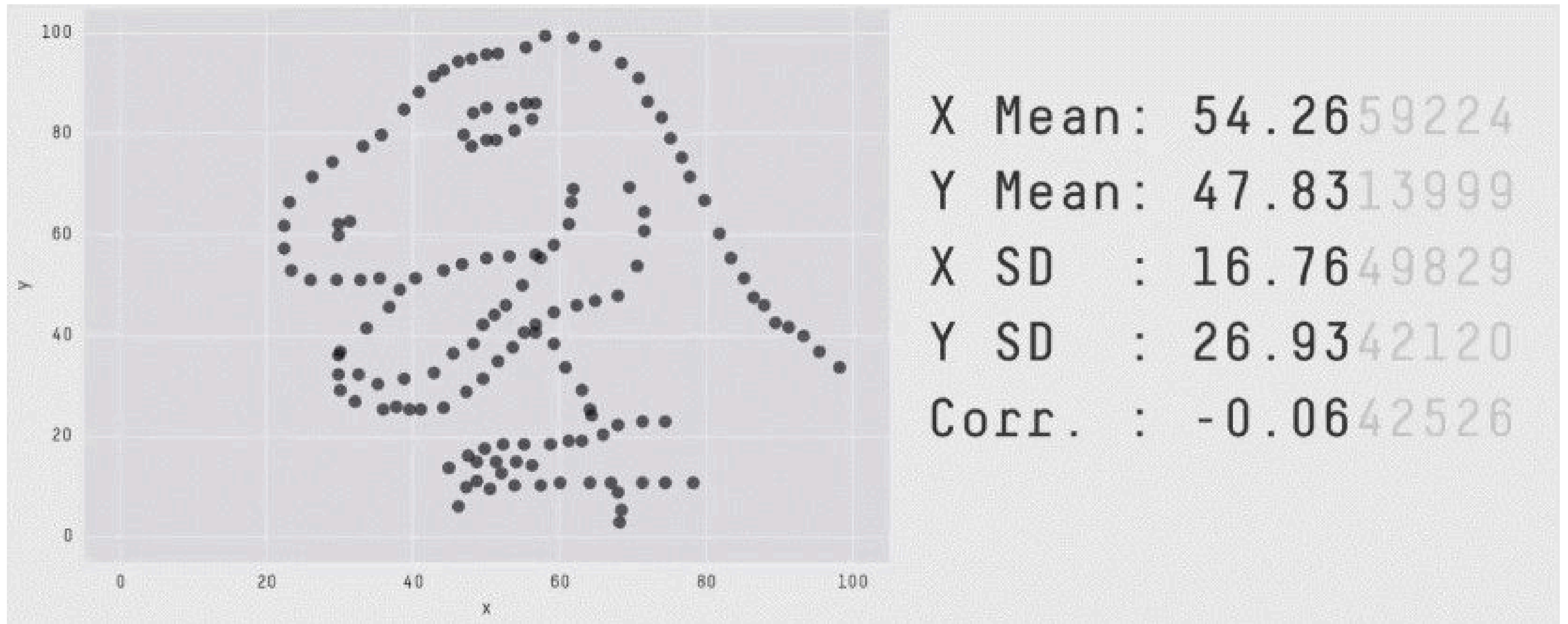


Set 3



Set 4





Same Stats, Different Graphs: Generating Datasets with Varied Appearance and Identical Statistics through Simulated Annealing (CHI 2017)

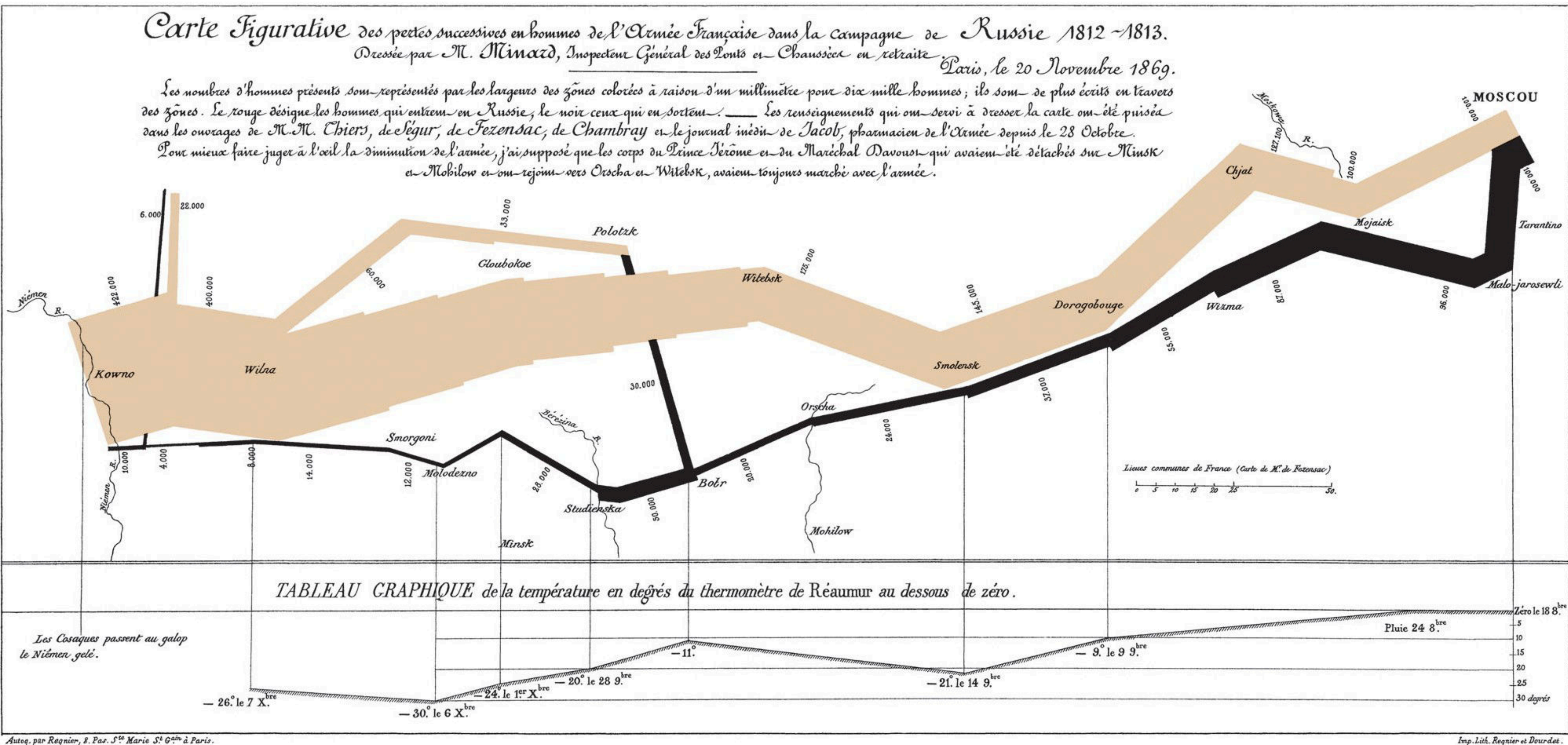
Justin Matejka, George Fitzmaurice

<https://www.autodeskresearch.com/sites/default/files/SameStats-DifferentGraphs.pdf>

Communicating information to others

Share and persuade

Telling a story with visualization



Communicate

Follow the lives of 21 boys who grew up in rich families ...

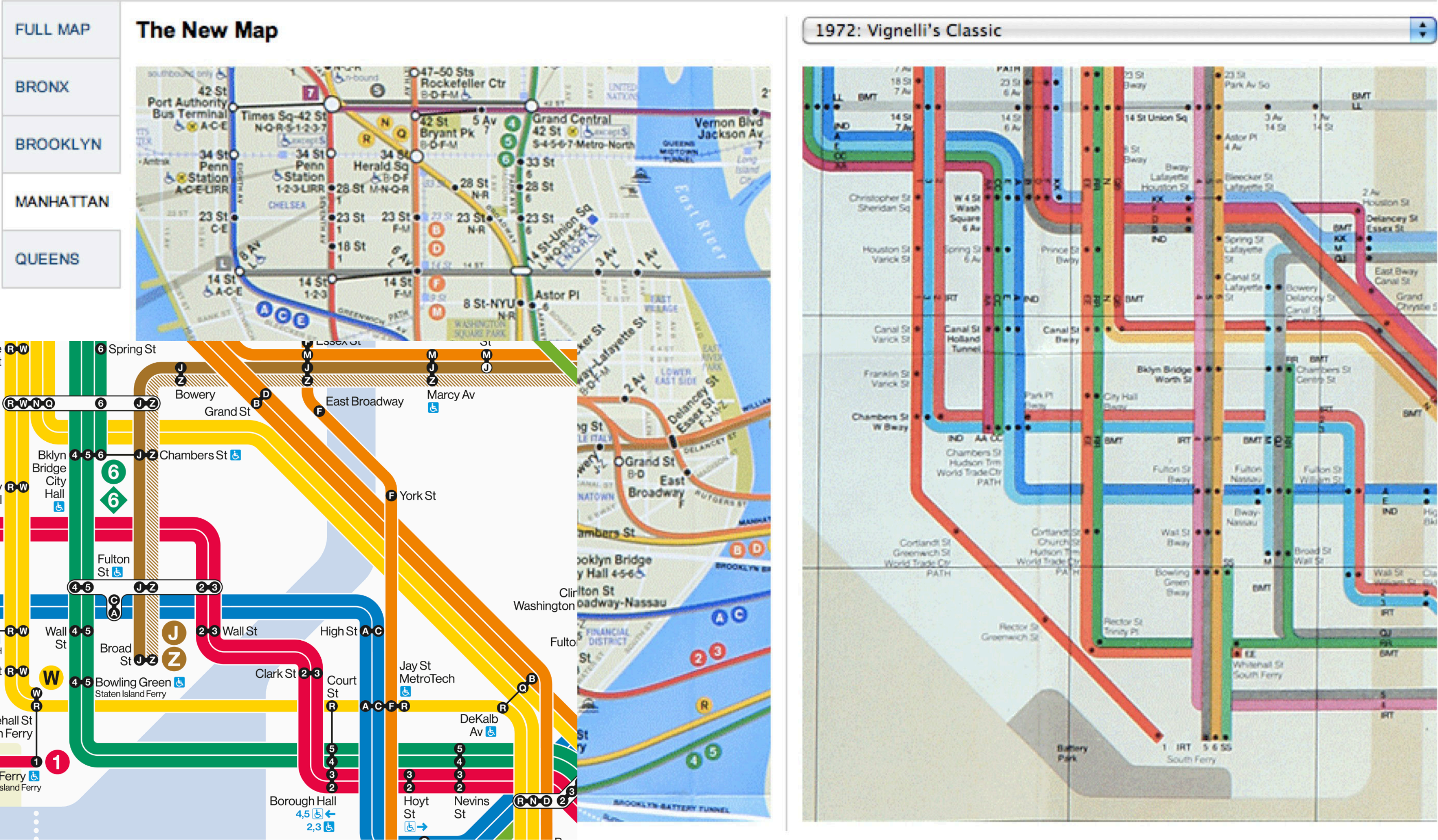
...and see where they end up as adults:



Adult outcomes reflect household incomes in 2014 and 2015.

An Overhaul of an Underground Icon

Next month, the Metropolitan Transportation Authority will unveil a resized, recolored and simplified edition of the well-known map, its first overhaul in more than a decade. [Related Article »](#)



Which map would you prefer?

- Open the class activity: smgroves.github.io/DS2003/getting-around-nyc
- We will all start at the same time, but you can study your map while we are waiting
- There will be a few questions to answer
- Whoever is the fastest will get a prize 😊

Good Data Visualization

... makes data accessible

... combines strengths of

humans and computers

... enables insight

... communicates

Project 1: Exploratory Data Analysis (EDA)

- We will discuss what this entails over the next few weeks
- Checkpoint #1: Topic and dataset picked: **by Sept 13 @ midnight**
- Checkpoint # 2: Questions to explore picked: **by Sept 27 @ midnight**
- Final **lightning talk** due: **Oct 11 @ midnight**
- Final **notebook** due: **Oct 14 @ midnight**
- For now, you should pick one of your interests (research or hobby!) and think of questions you have about that interest

Coffee:

- Is Charlottesville coffee overpriced?
- What makes for a good cup of coffee? (CQI)
- Do coffee consumers tend to prefer light or dark roast?
- How does the local vs global economy affect coffee prices and sales?

Public transit:

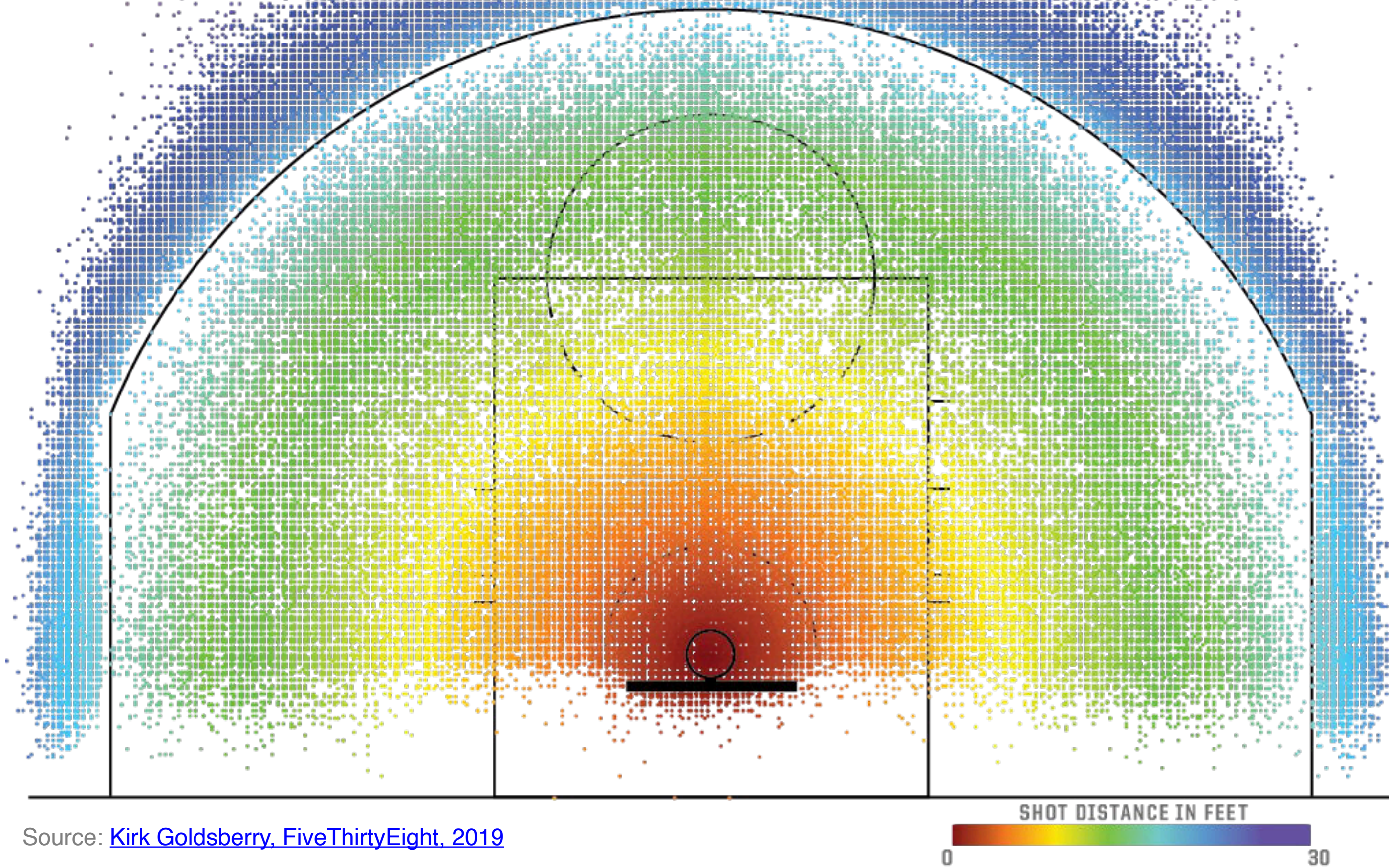
- How often are Charlottesville buses late?
- How much overlap is there between UTS and CAT?
- How has ridership changed since COVID? Was the decision to make rides free a good one?

Exit ticket and quizzes

- Exit tickets will sometimes be used on Monday, graded for participation, NOT right answers! They are also anonymous
- They will help inform if I need to go over anything again that was confusing, and to make sure you are understanding the content and encoding knowledge!
- There is **NOT** a quiz next class; weekly quizzes will start September 9 (next week)
 - First 10 min in class
 - They will cover anything from the previous lectures; if you are coming to class and completing the assignments you will do fine 😊
 - Lowest two quiz grades are dropped out of 10 total

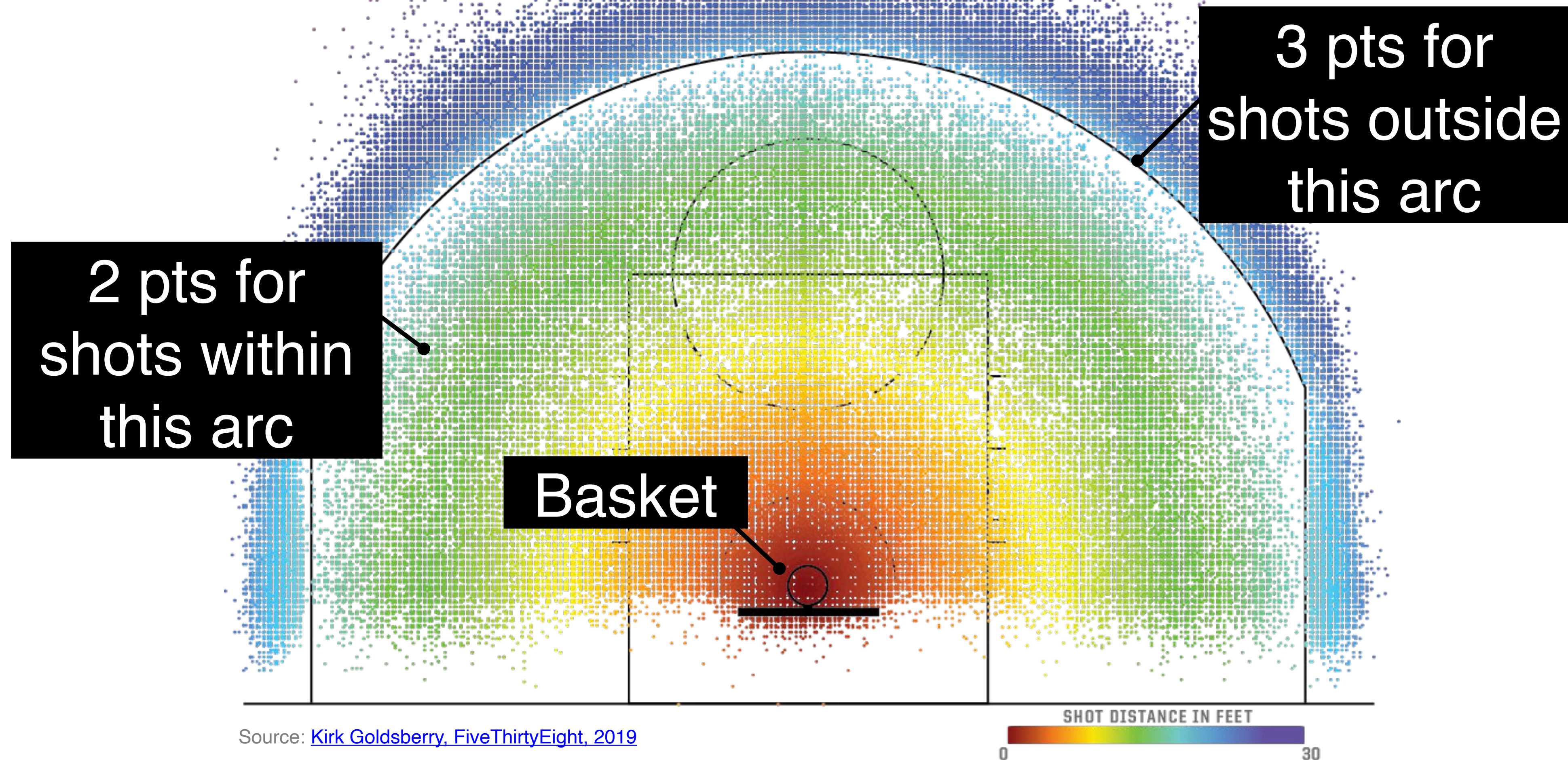
**Is the game of
basketball broken?**

NBA SHOT LOCATIONS 2014-15



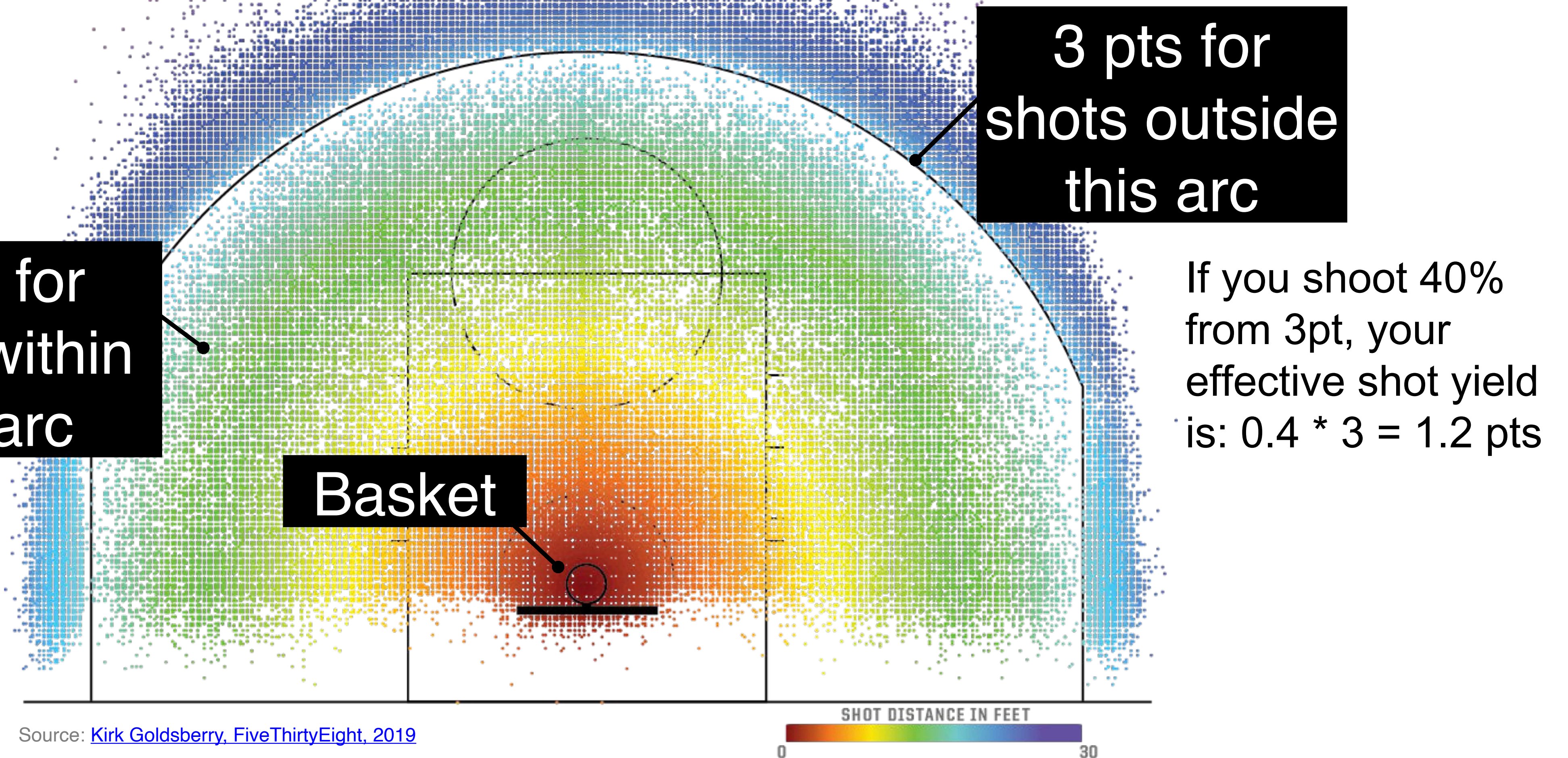
Source: [Kirk Goldsberry, FiveThirtyEight, 2019](#)

NBA SHOT LOCATIONS 2014-15



Source: [Kirk Goldsberry, FiveThirtyEight, 2019](#)

NBA SHOT LOCATIONS 2014-15



Source: [Kirk Goldsberry, FiveThirtyEight, 2019](#)

If you shoot 50% from 2pt, your effective shot yield is: $0.5 * 2 = 1$ pt

Pro Basketball

Search All NYTimes.com

 Go

Where the Heat and the Thunder Hit Their Shots

Shooting patterns for the players on the Miami Heat and the Oklahoma City Thunder reveal where they are most dangerous on the court. Below, compare each player's strengths using court maps and analysis by Kirk Goldsberry, a geography professor at Michigan State. [Related Article »](#)

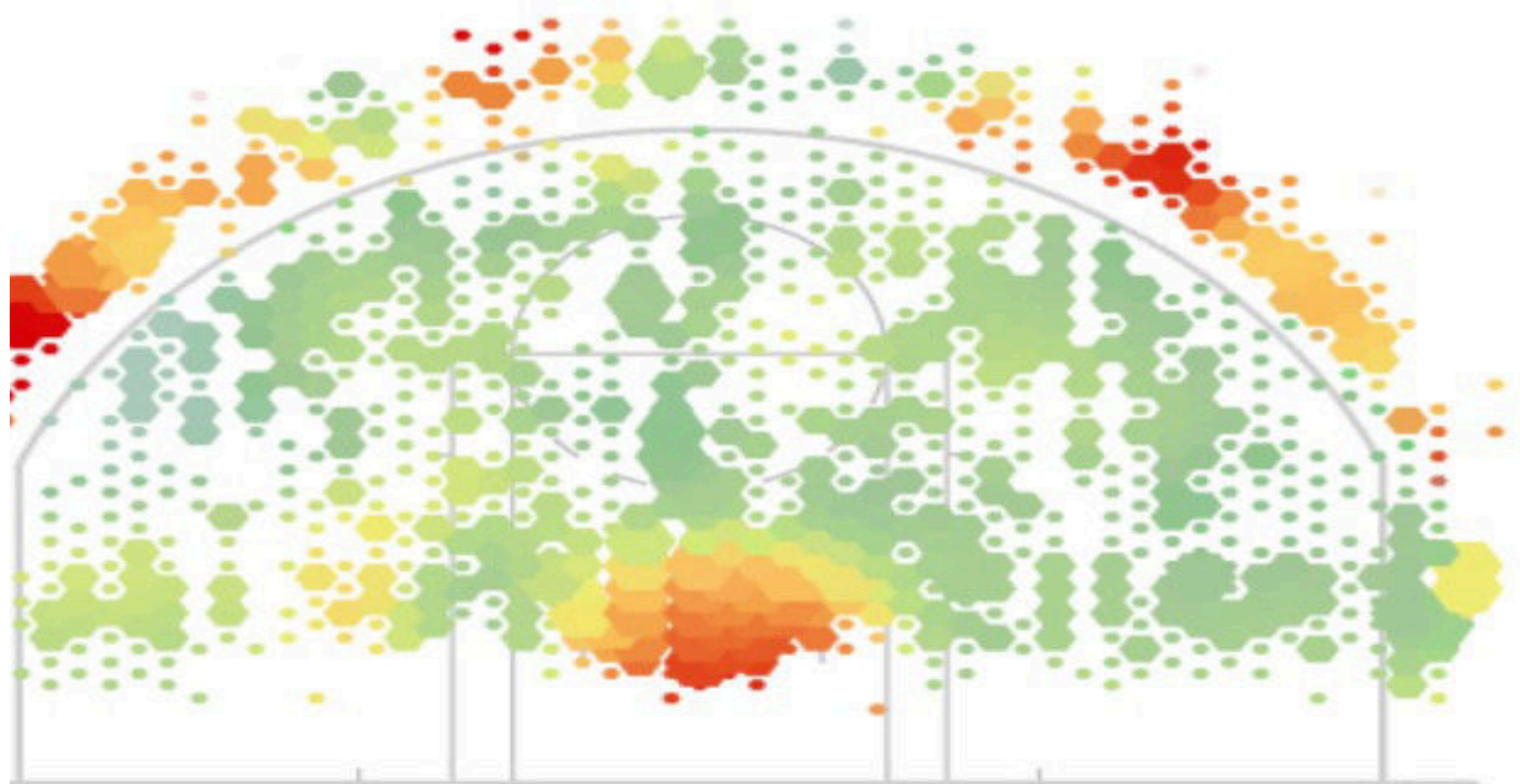
All Shots
3-Pointers
Midrange
Close Range

Number of attempts
Low ○ ○ ○ High

Points per region
Low High

Miami Heat

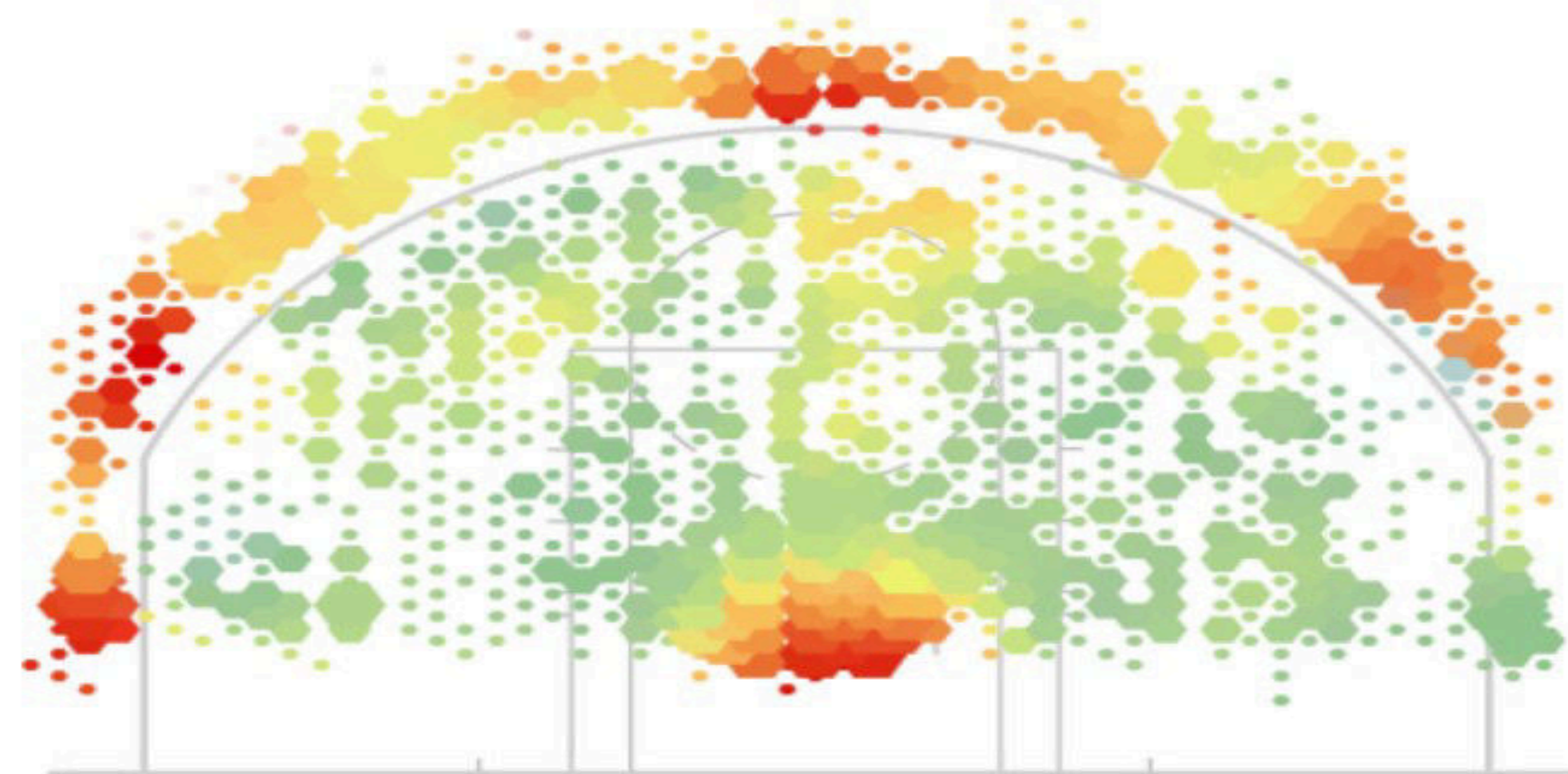
TOTAL SHOTS **5,209** | POINTS PER SHOT **1.01** | F.G. PERCENT **47%**



The Heat rely on player positioning to create isolation plays for LeBron James and Dwyane Wade, often on the left side. The Heat take many fewer 3-point shots than the Thunder.

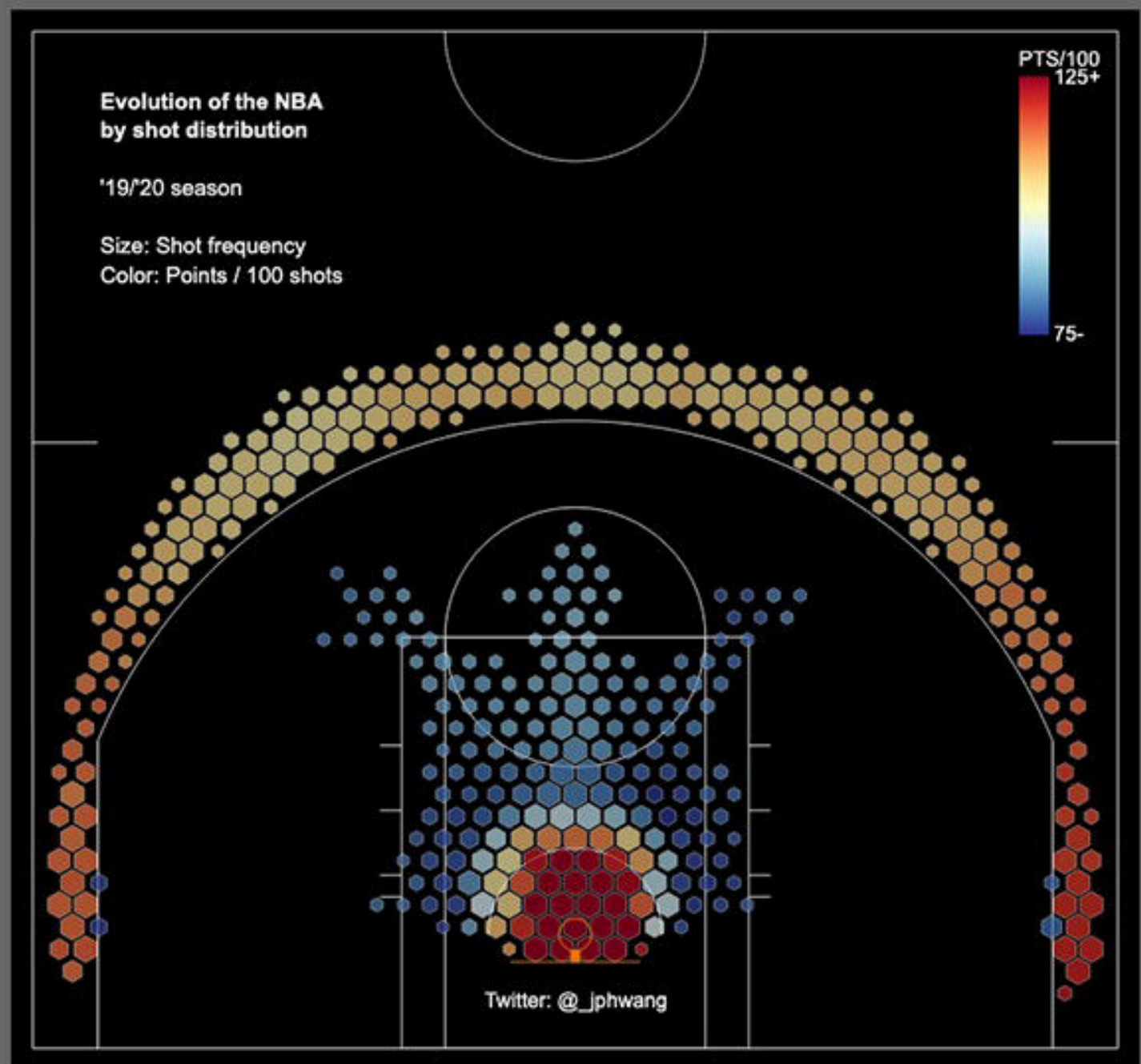
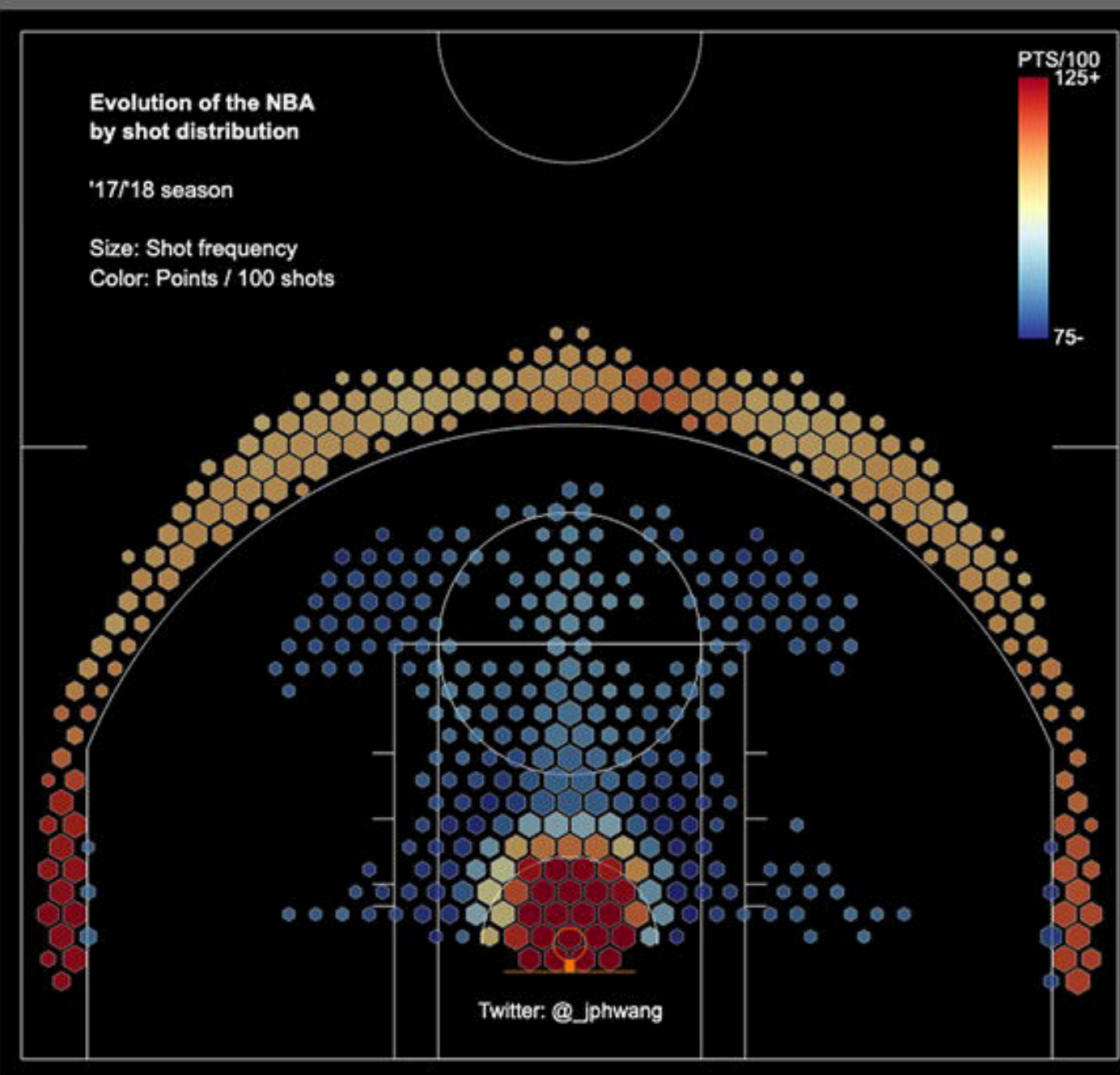
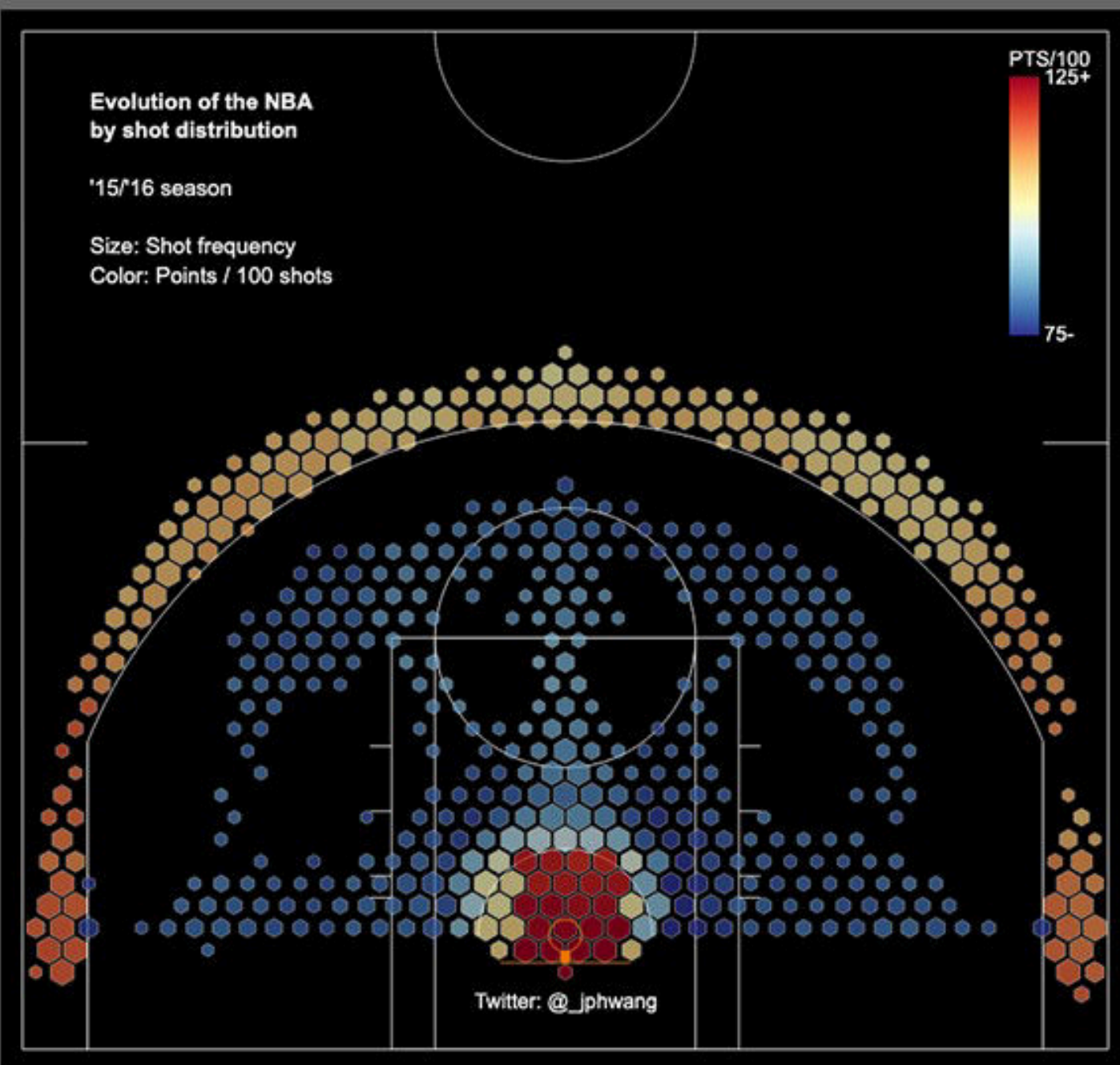
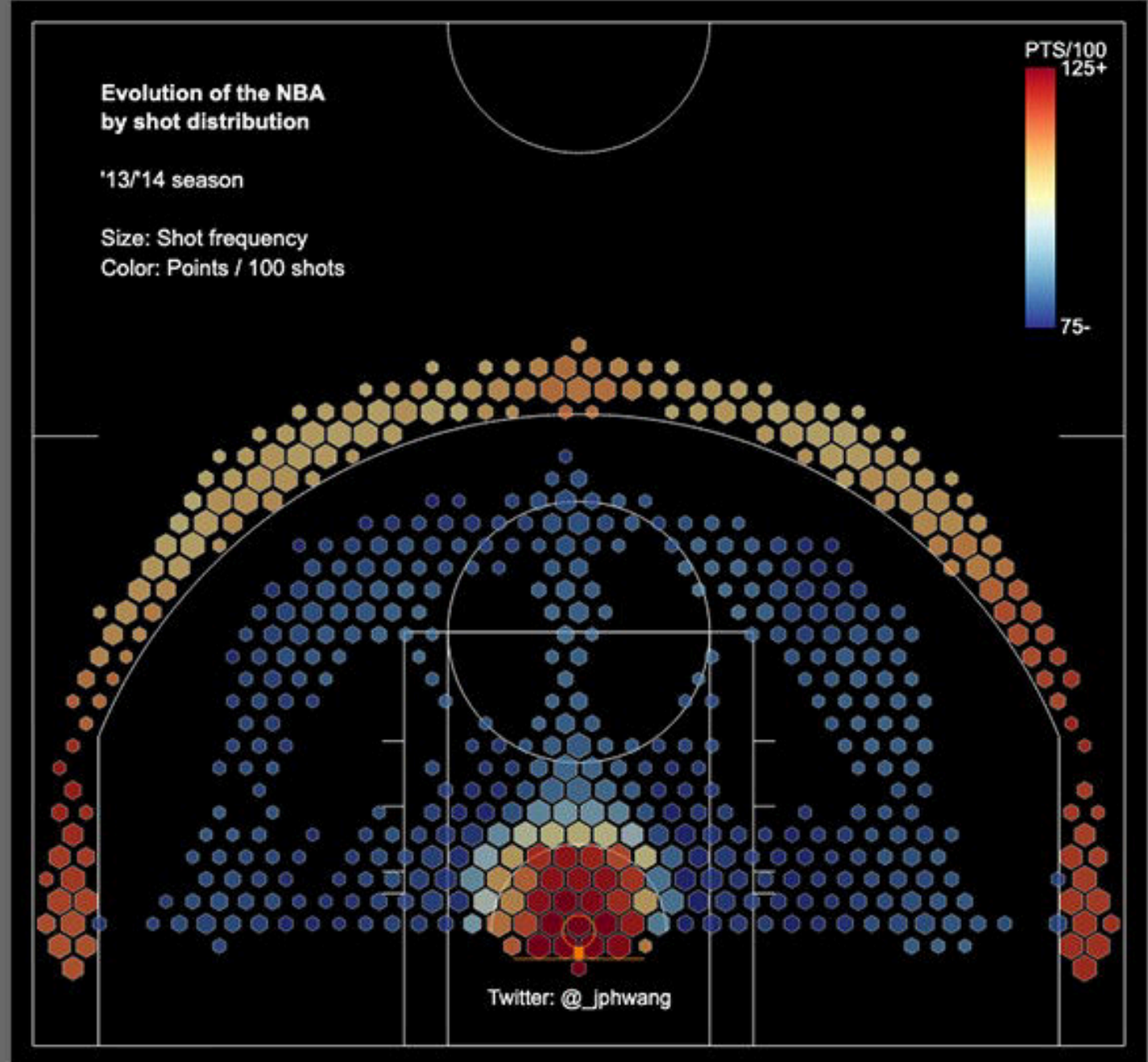
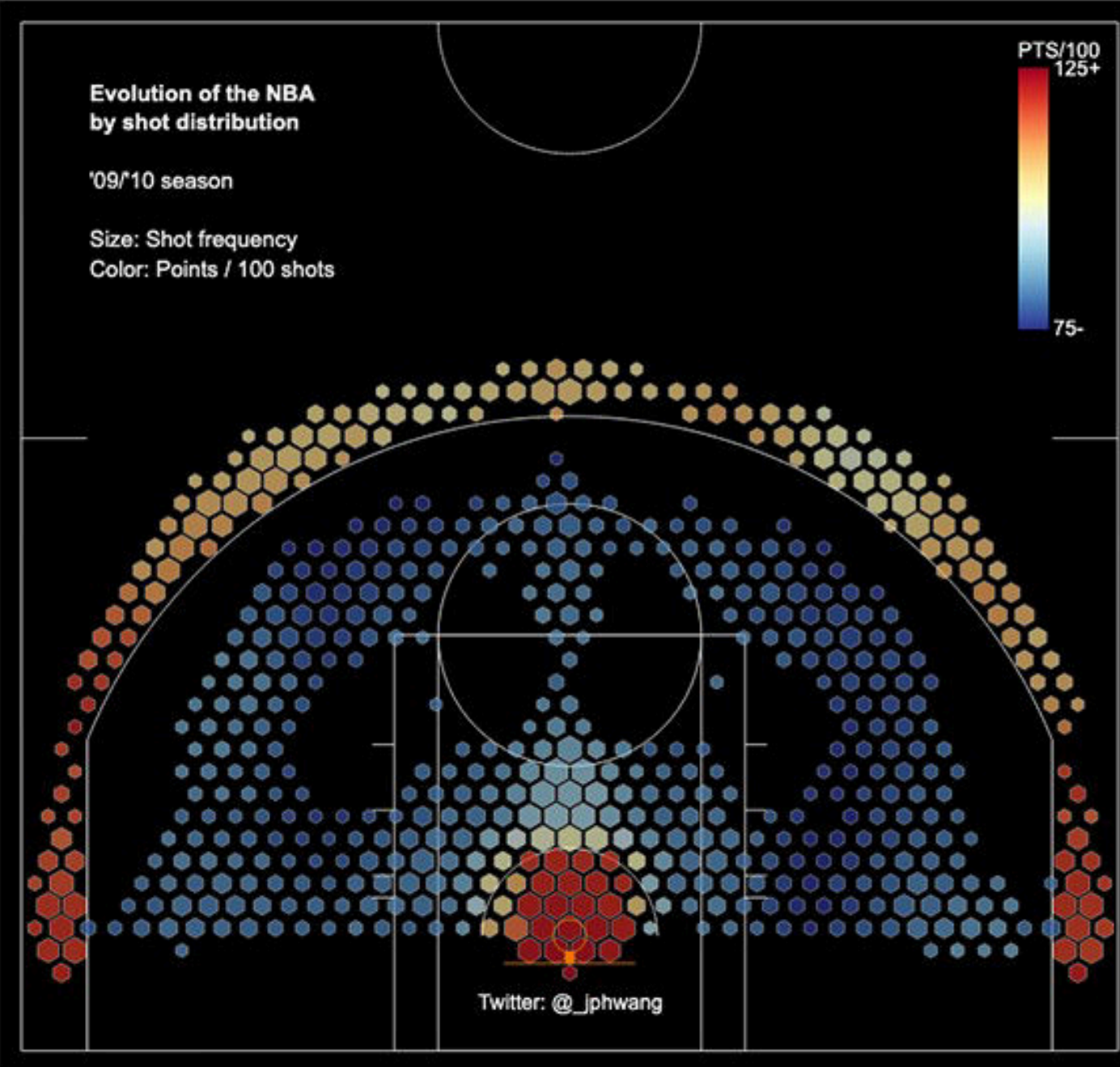
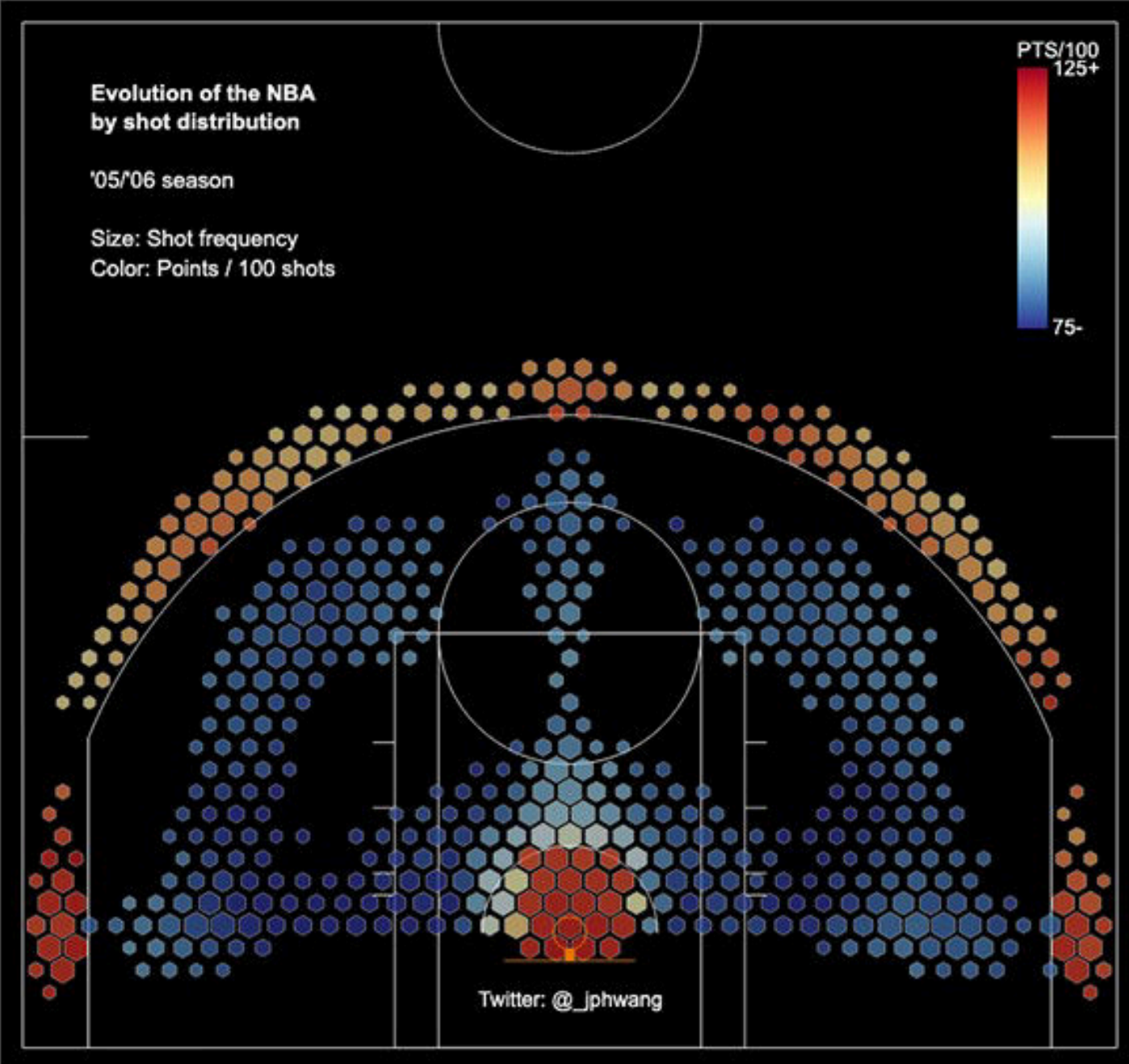
Oklahoma City Thunder

TOTAL SHOTS **5,228** | POINTS PER SHOT **1.03** | F.G. PERCENT **47.1%**



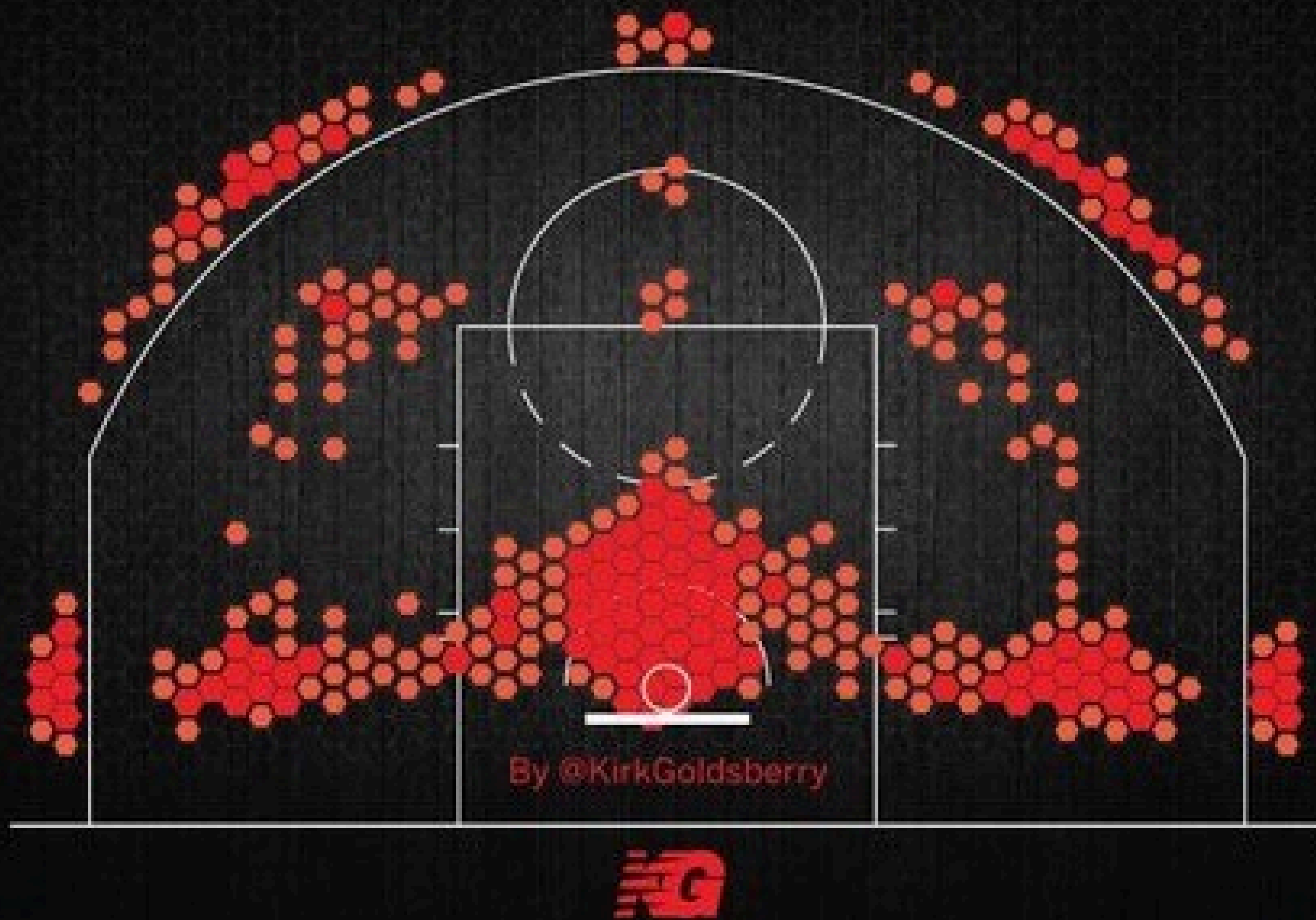
The Thunder are effective from almost any area on the court and shoot many more 3-point shots than the league average. Kevin Durant and James Harden are potent from the top of the arc.

Source: [NYTimes, 2012](#)



Top 200 shot locations in the NBA, 2001-02 versus 2019-20

2001-02

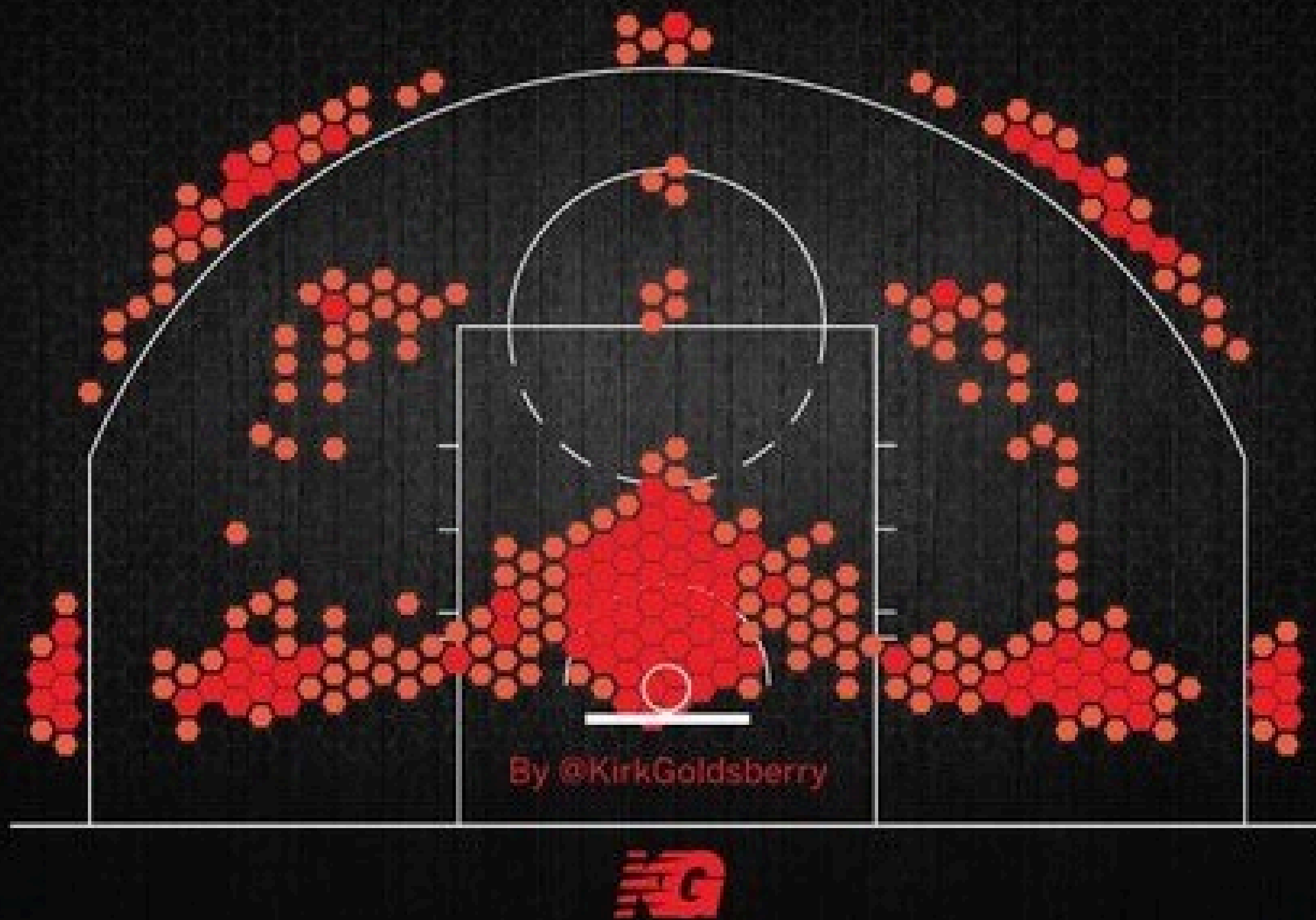


2019-20

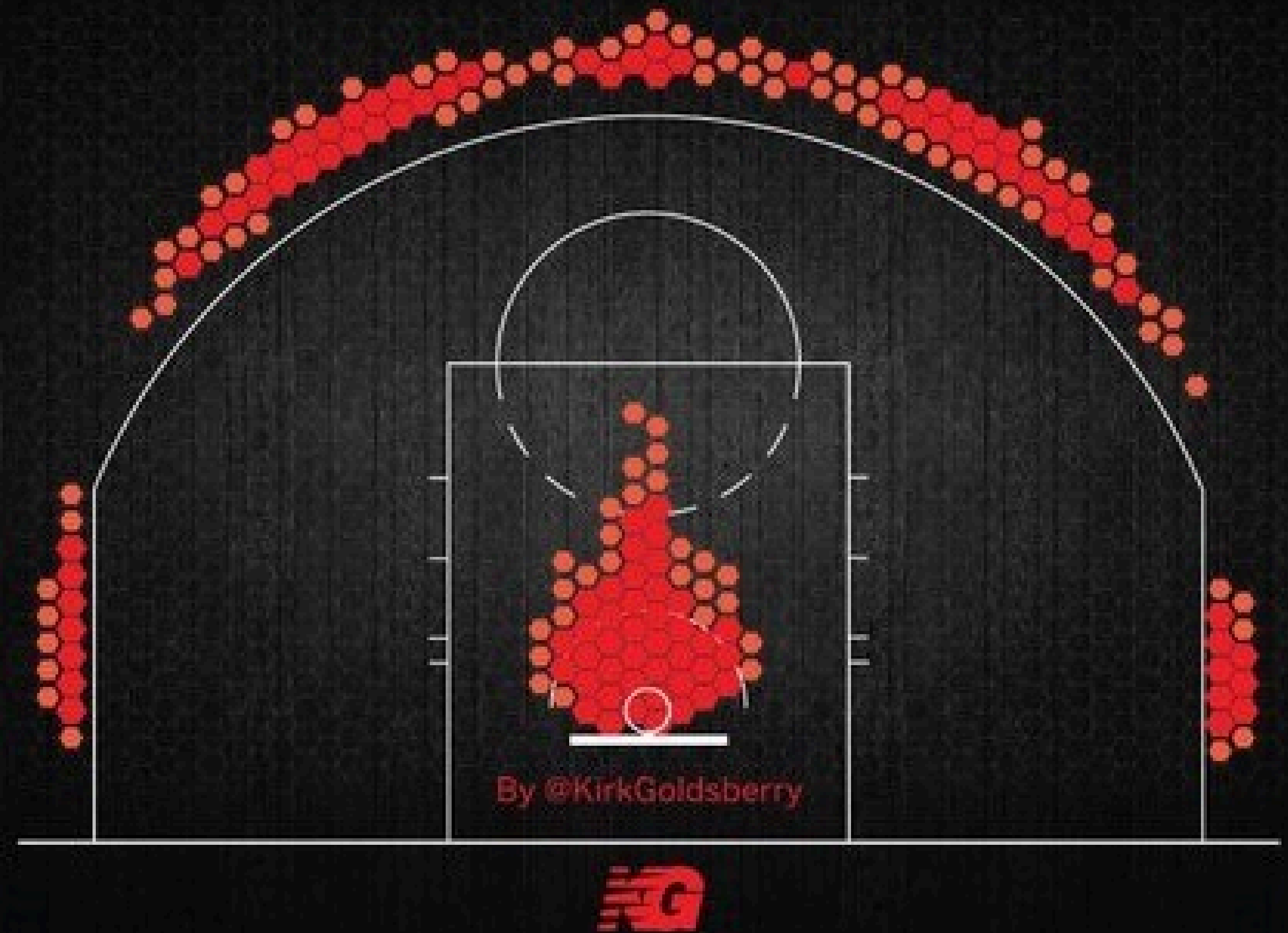


Top 200 shot locations in the NBA, 2001-02 versus 2019-20

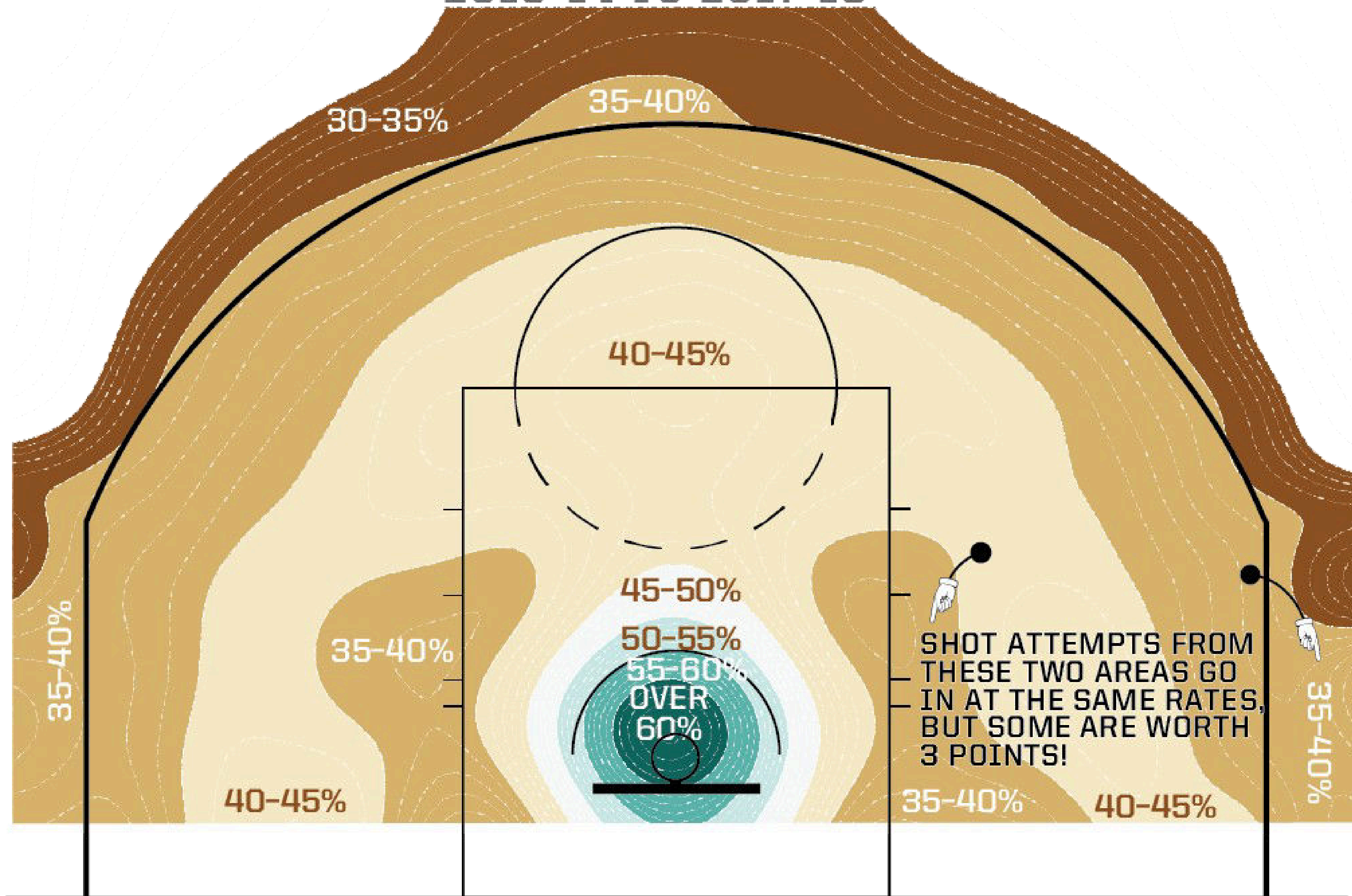
2001-02



2019-20

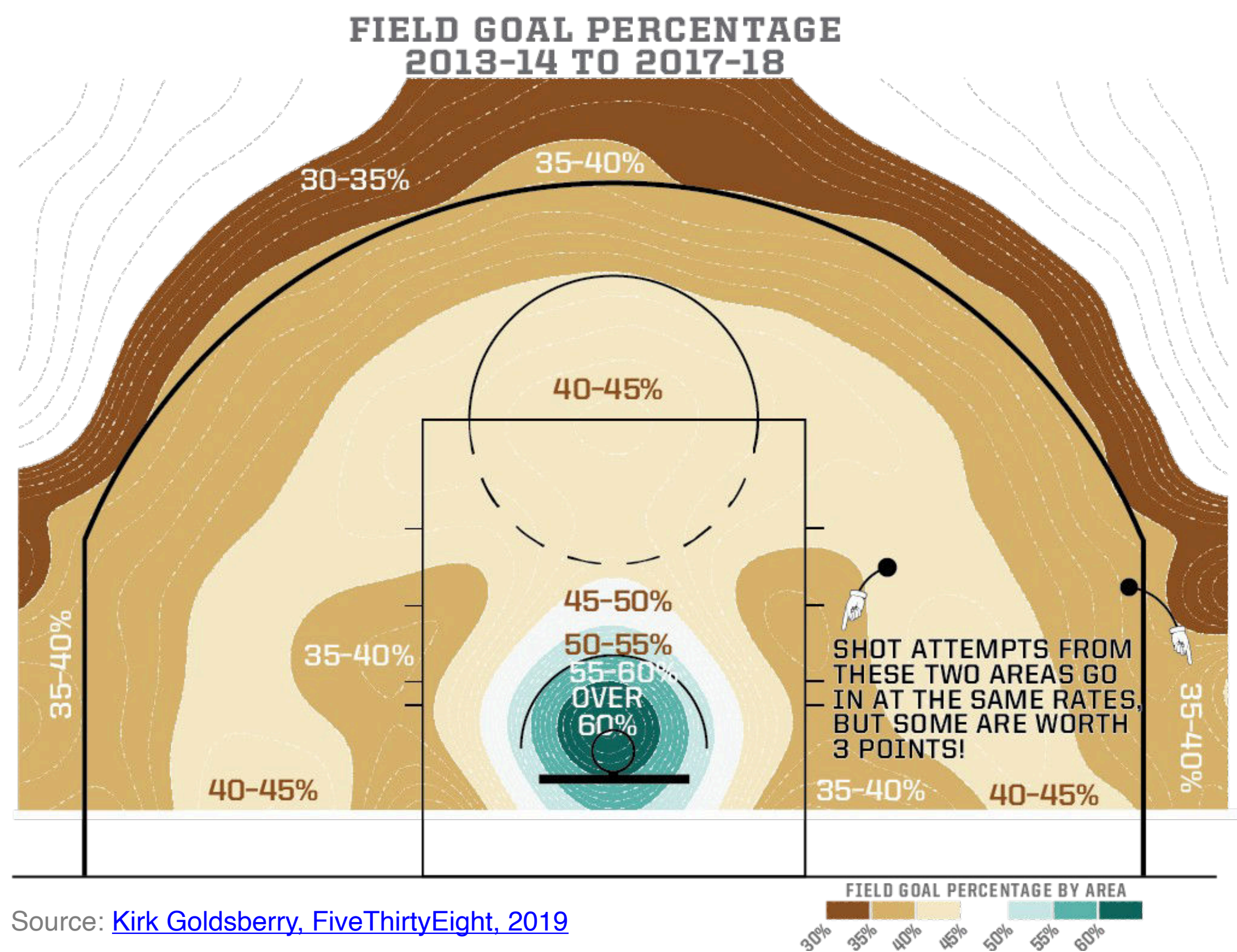


FIELD GOAL PERCENTAGE 2013-14 TO 2017-18

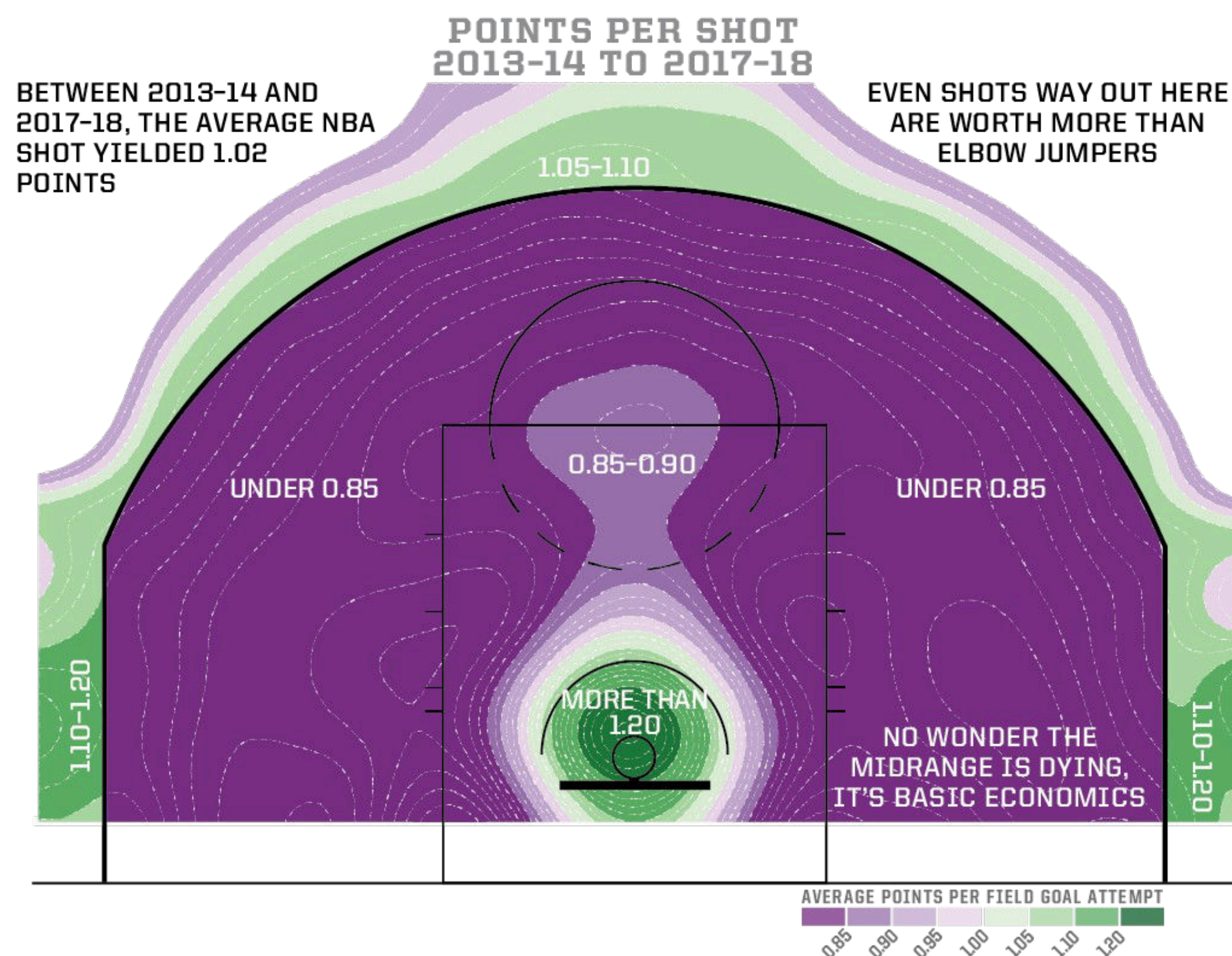


Source: [Kirk Goldsberry, FiveThirtyEight, 2019](#)



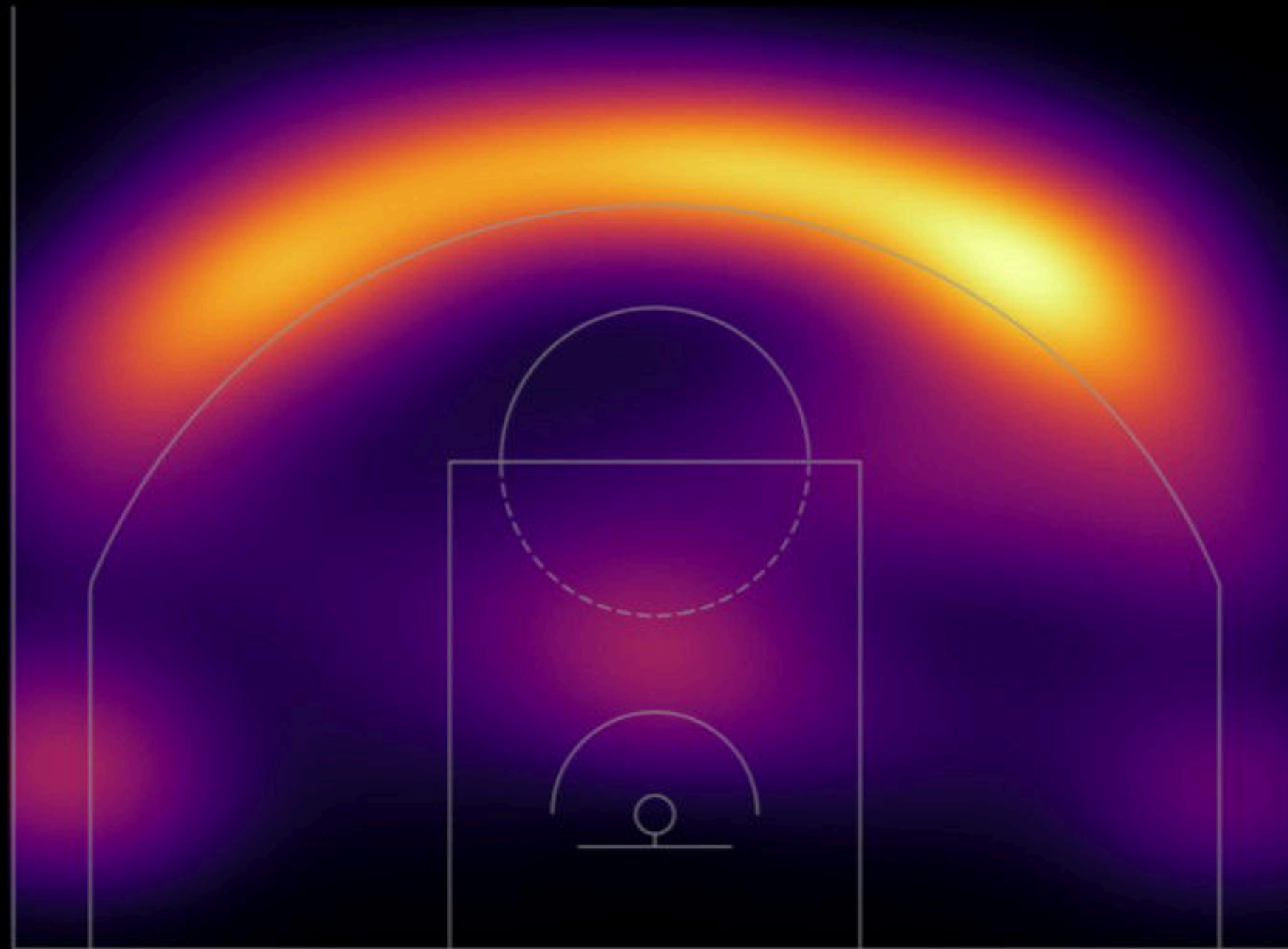


Source: [Kirk Goldsberry, FiveThirtyEight, 2019](#)



Stephen Curry

2015-16 Regular Season
Golden State Warriors



Shot frequency
lower higher

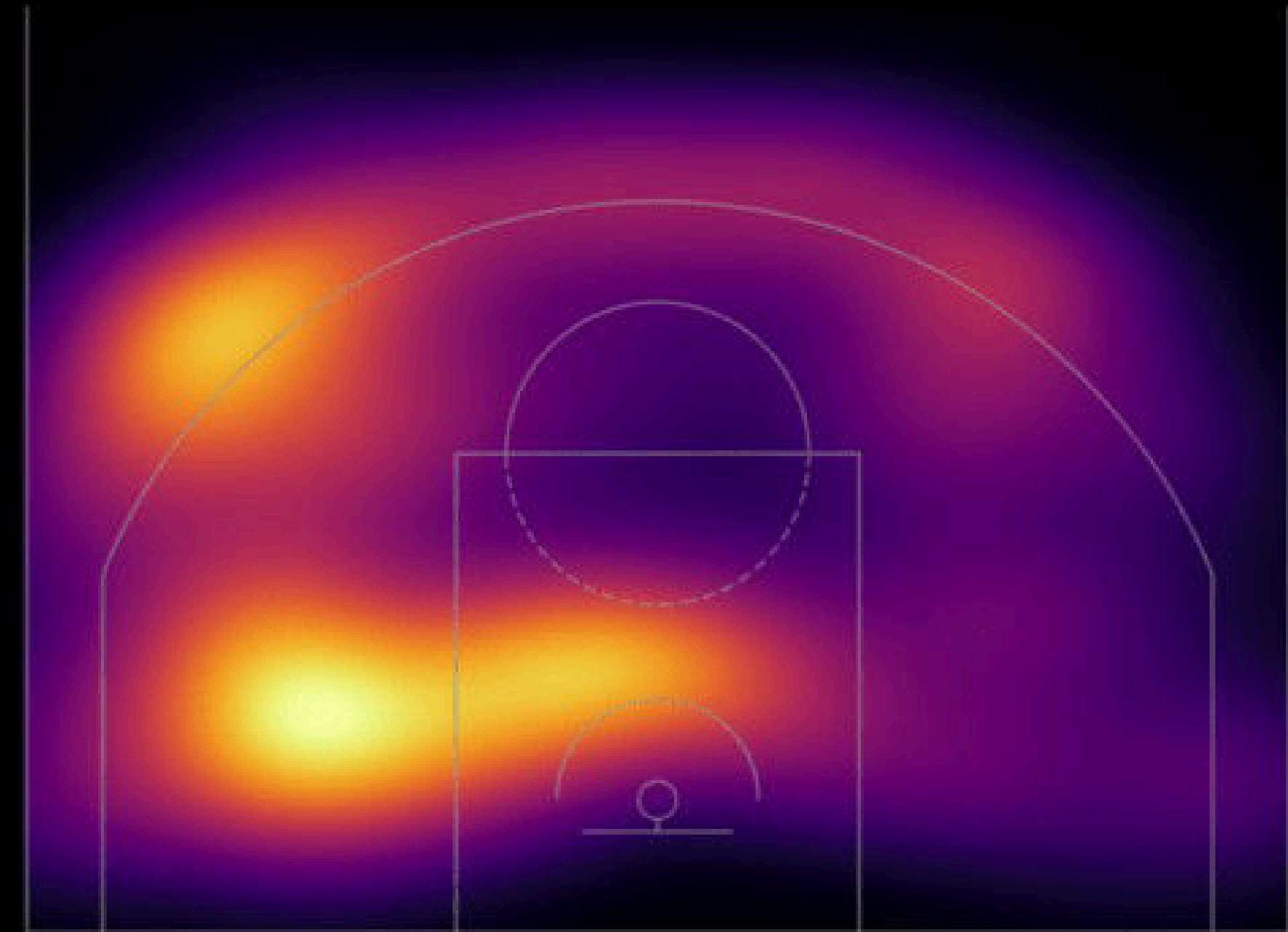
Shot Filters Applied

Zone: Above the Break 3, Left Corner 3, Right Corner 3, Mid-Range, In The Paint (Non-RA)

Data via stats.nba.com
toddschneider.com/ballr

LeBron James

2015-16 Regular Season
Cleveland Cavaliers



Shot frequency
lower higher

Shot Filters Applied

Zone: Above the Break 3, Left Corner 3, Right Corner 3, Mid-Range, In The Paint (Non-RA)

Data via stats.nba.com
toddschneider.com/ballr